

**Factors Associated with Longevity in Companion Dogs:  
Initial Findings from the Dog Aging Project**

**Supplemental Figure S15**

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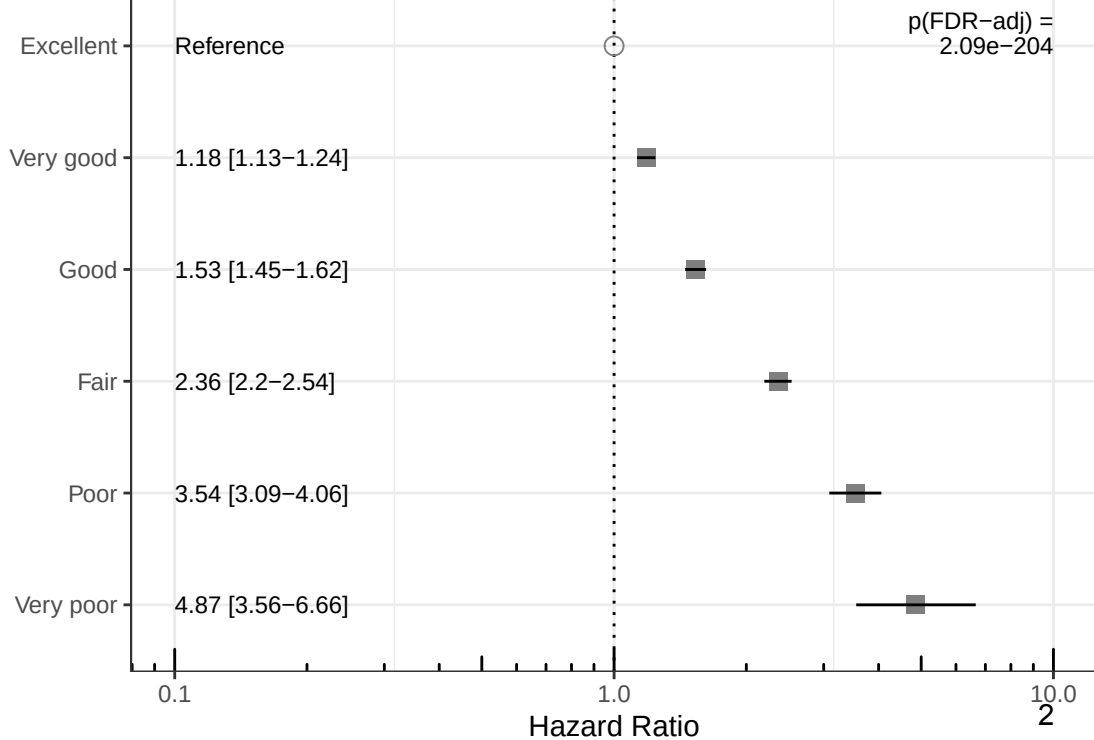
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\*Corresponding Author. Email: [wchiu@tamu.edu](mailto:wchiu@tamu.edu)

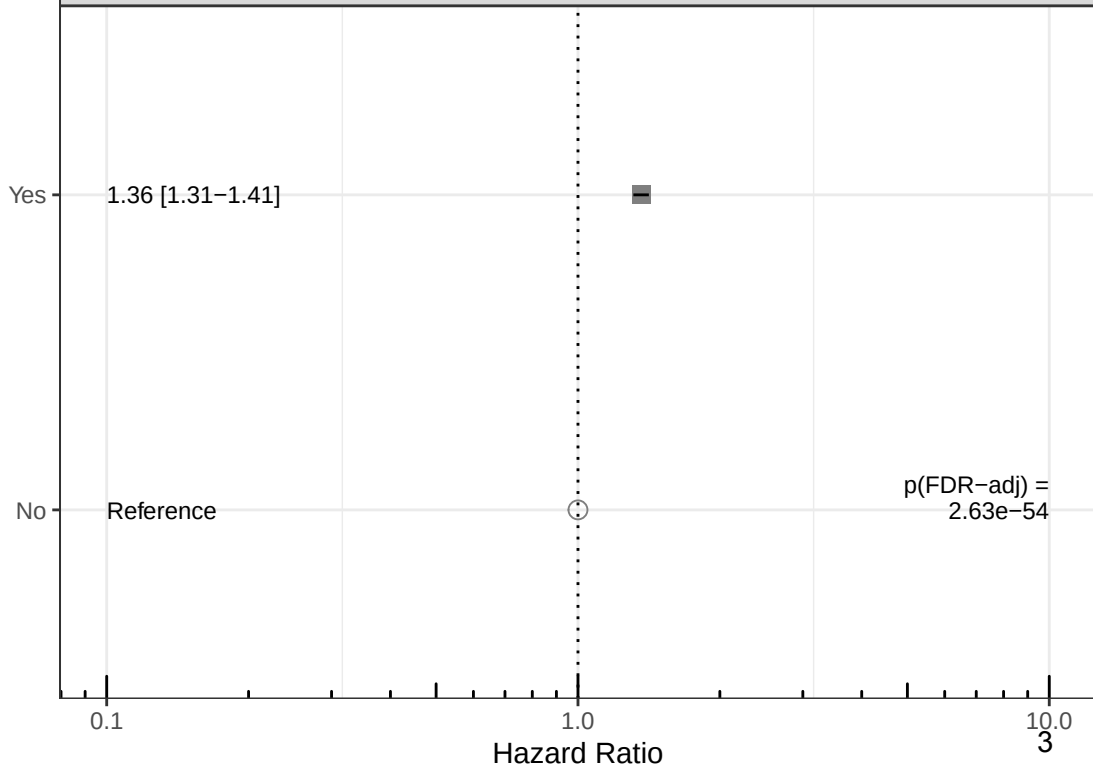
ORCID ID: <https://orcid.org/0000-0002-7575-2368>

**Supplemental Figure S15.** Forest plots of hazard ratios and FDR-corrected p-values for categorical HLES variables from SWAS. Each panel shows the variable abbreviation and corresponding survey question in the HLES. Each hazard ratio (and confidence interval) corresponds to the increase or decrease relative to the reference category, which is defined by the category with the most frequent response.

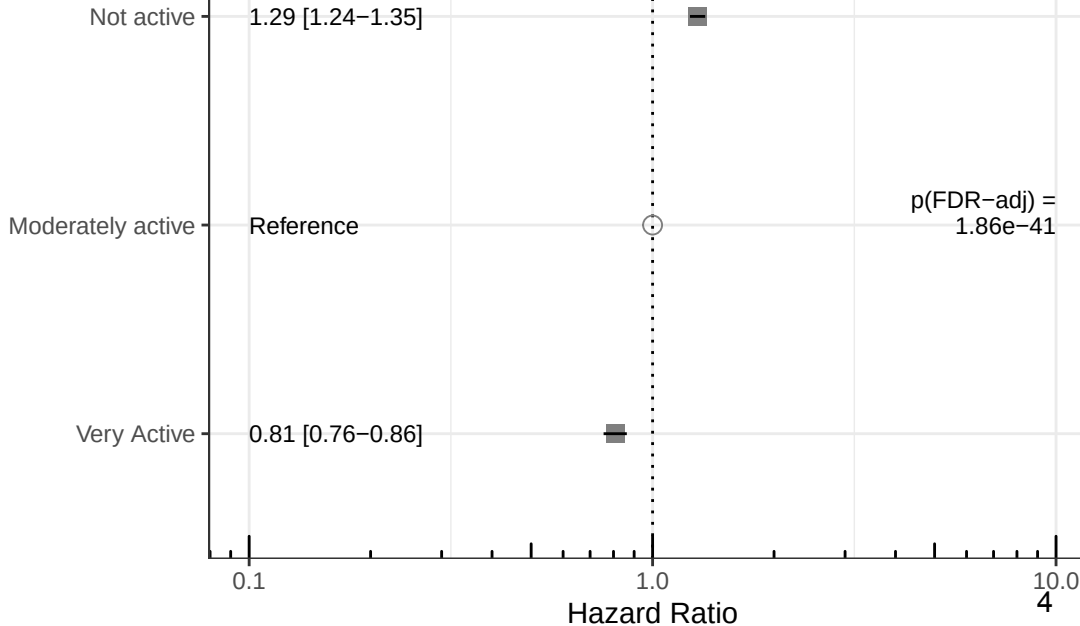
hs\_general\_health  
Would you say that in general your dog's health is:

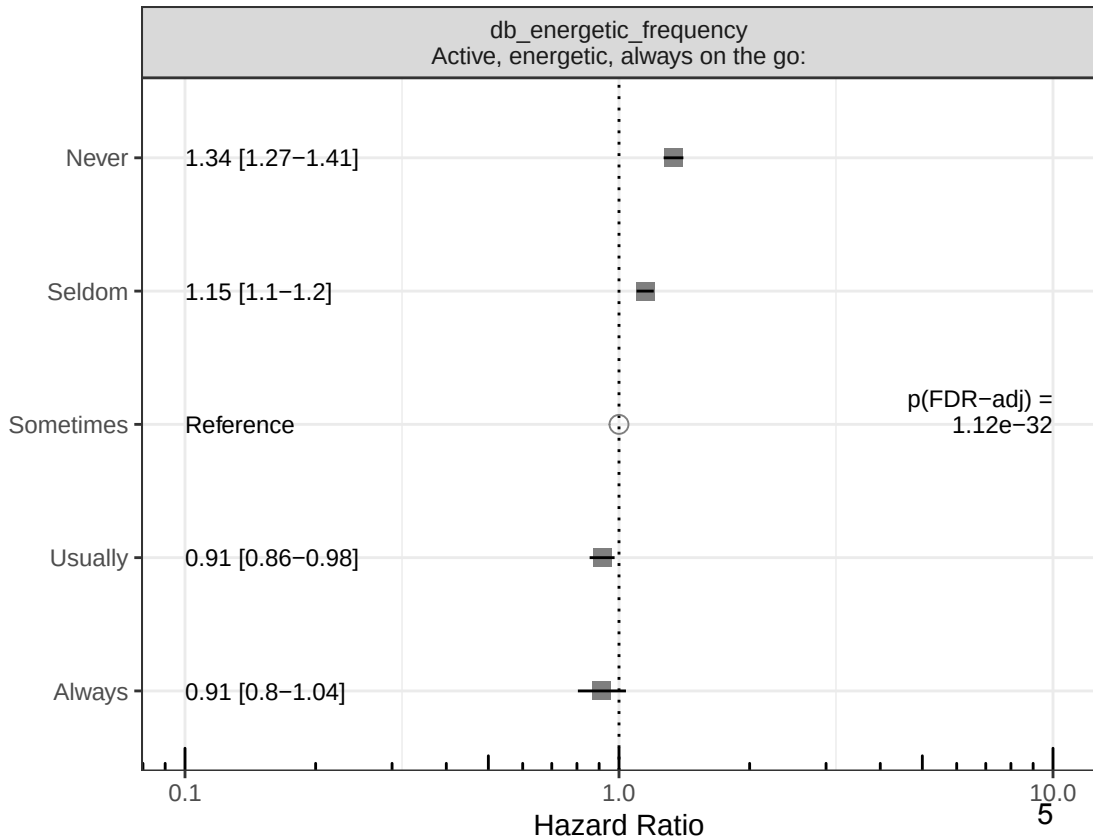


hs\_chronic\_condition\_present  
Does your dog have any ongoing medical conditions?

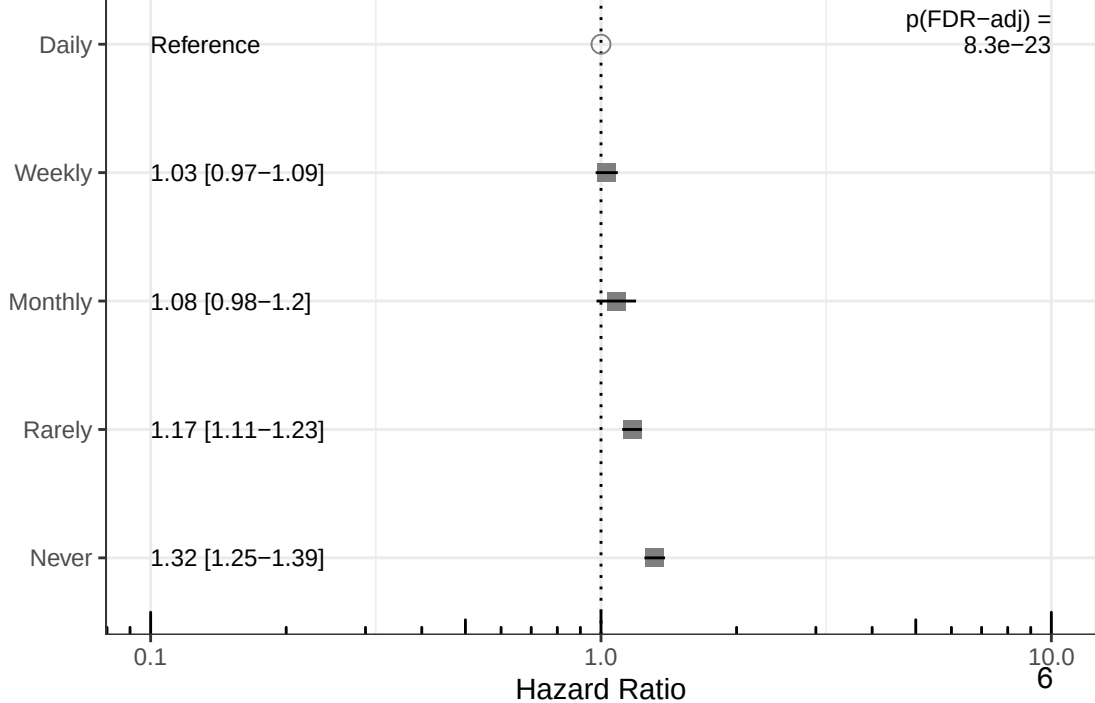


pa\_activity\_level  
Please choose the best description of your dog's lifestyle over the  
past year:

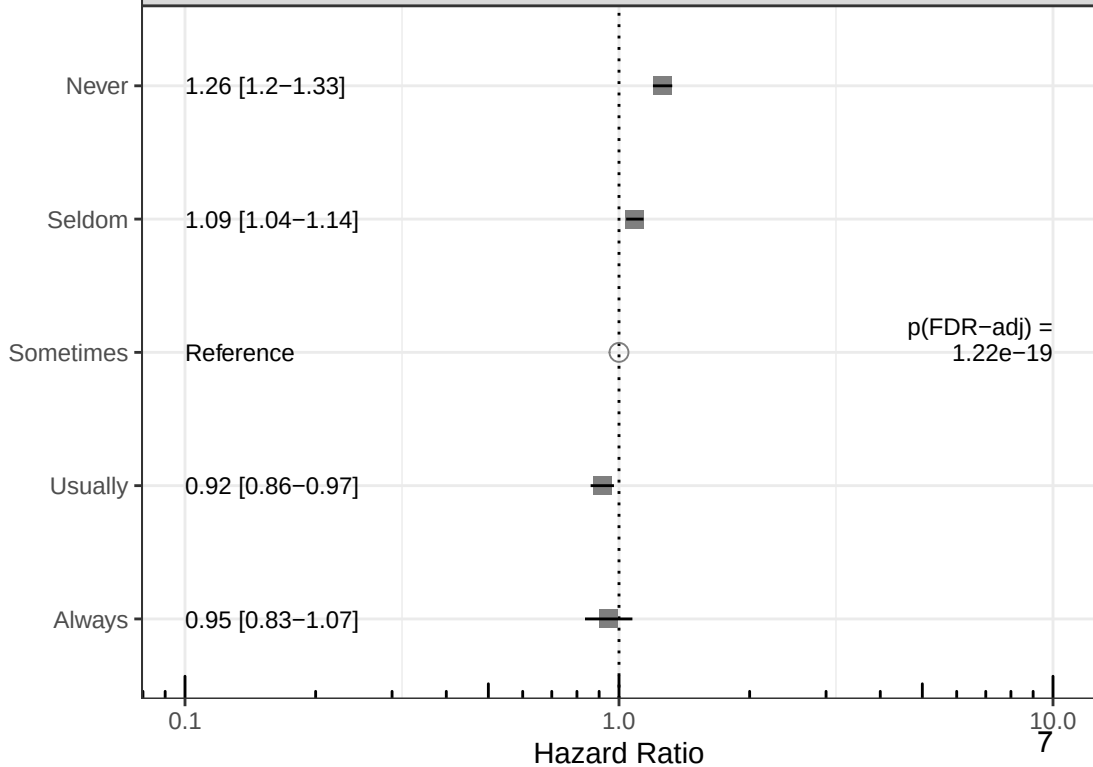




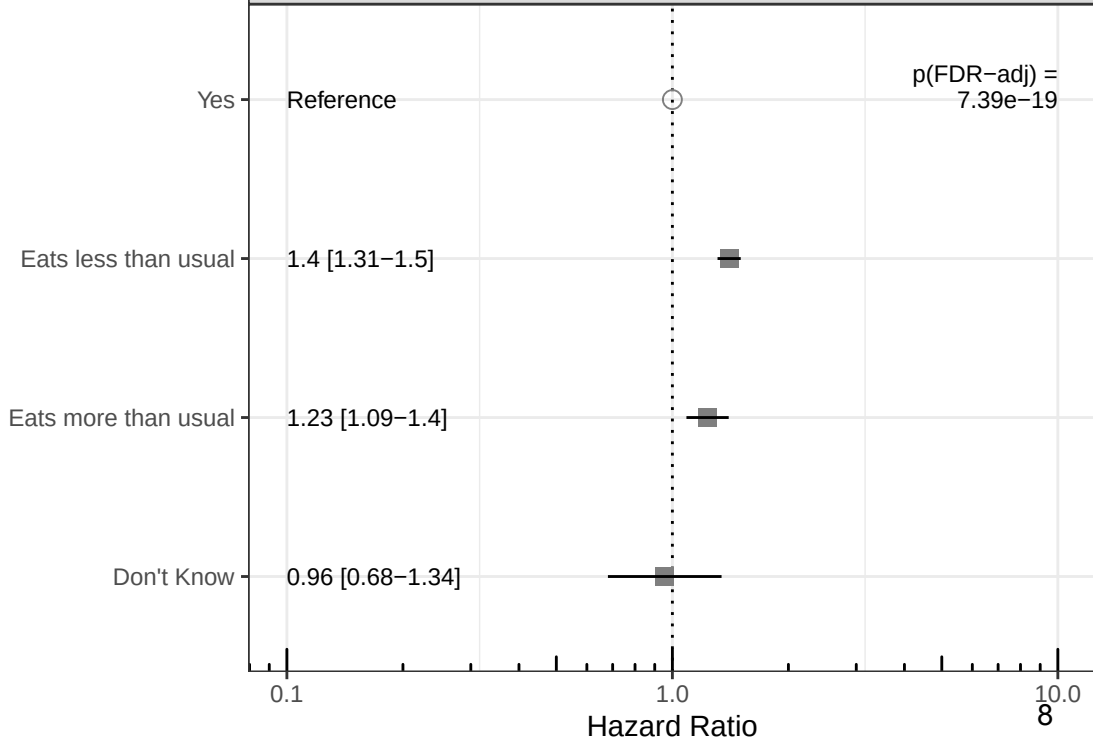
pa\_physical\_games\_frequency  
How often does your dog fetch items or play other games (such as  
Frisbee) that involve physical activity?



db\_playful\_frequency  
Playful, puppyish, boisterous:



df\_appetite\_change\_last\_year  
In the past 12 months, has your dog's appetite stayed about the same?  
How has your dog's appetite changed in the past 12 months?





pa\_swim  
Does your dog go swimming?

Yes  
0.81 [0.78-0.85]

No  
Reference

p(FDR-adj) =  
8.67e-18

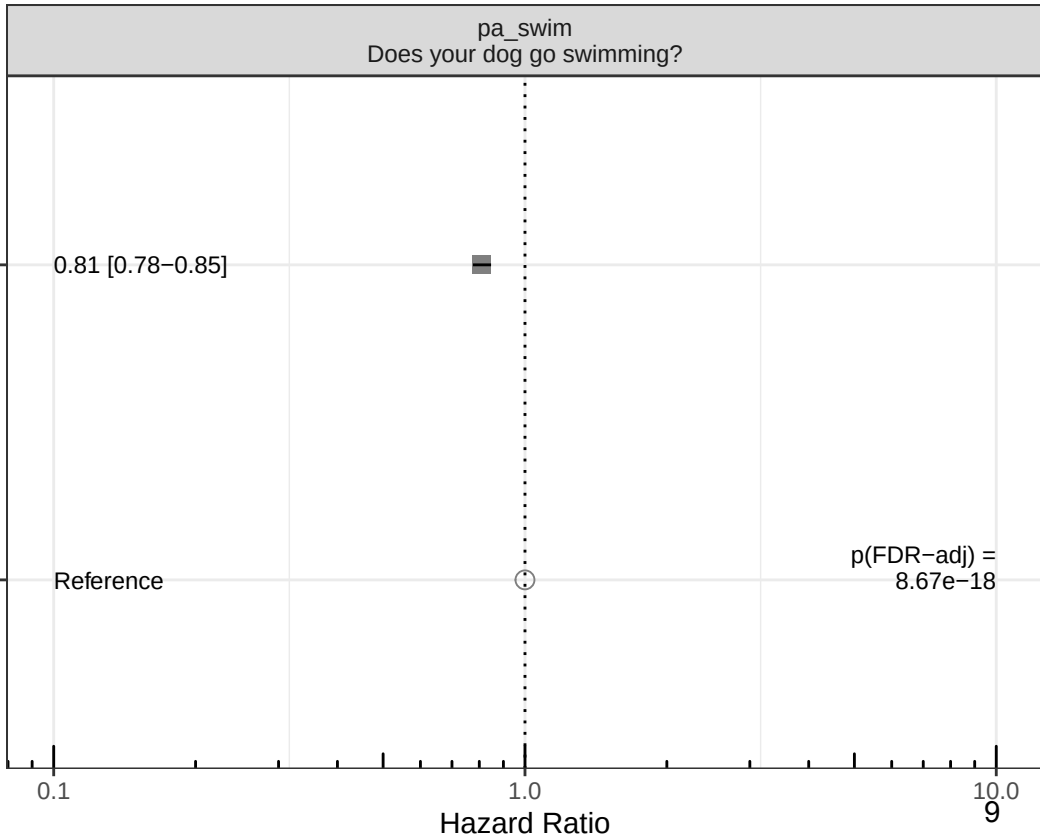
Hazard Ratio

0.1

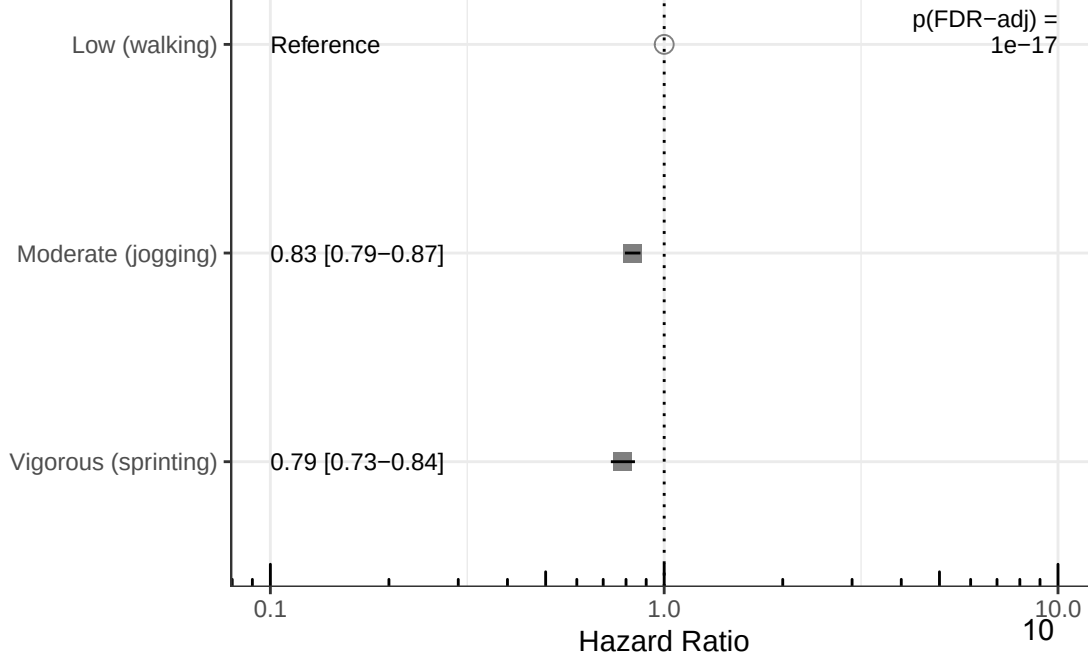
1.0

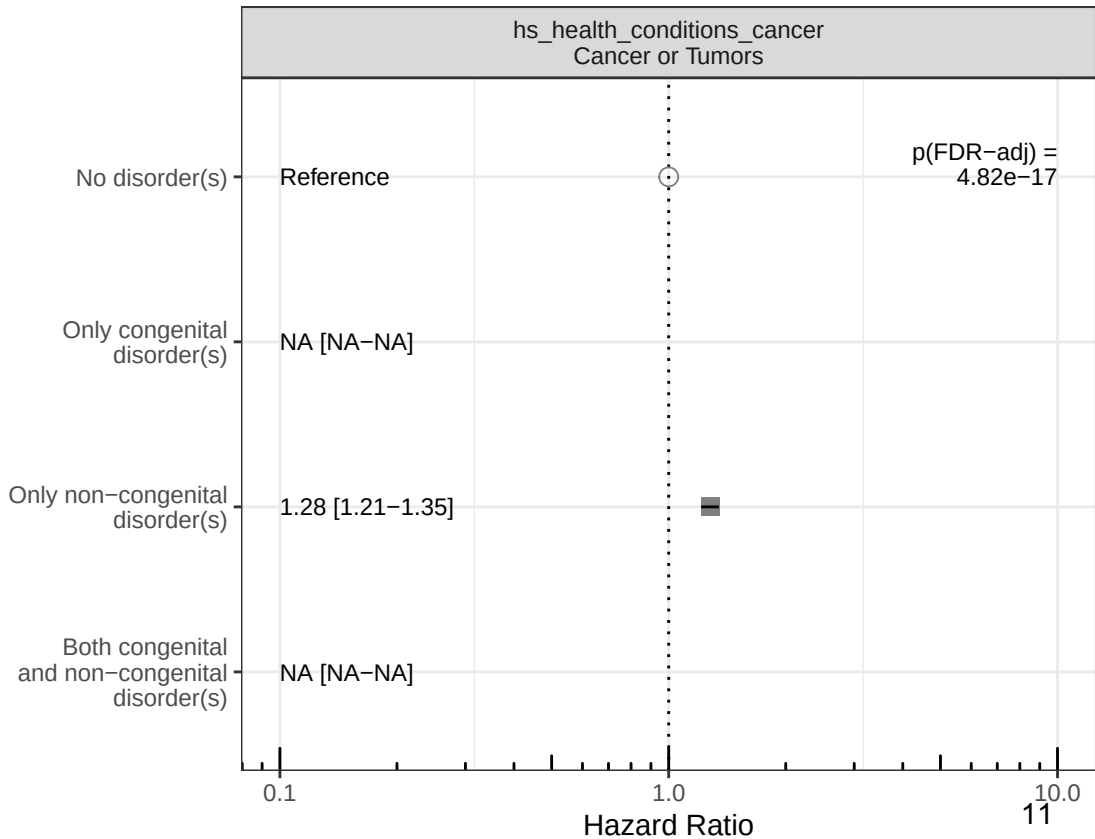
10.0

9



pa\_avg\_activity\_intensity  
Over the past year, when your dog is being active what is the average intensity level of that activity?





hs\_new\_condition\_diagnosed\_recently  
In the past three months, has your dog been diagnosed with any new conditions?

Yes  
1.24 [1.18–1.29]

No  
Reference

p(FDR-adj) =  
6.47e-17

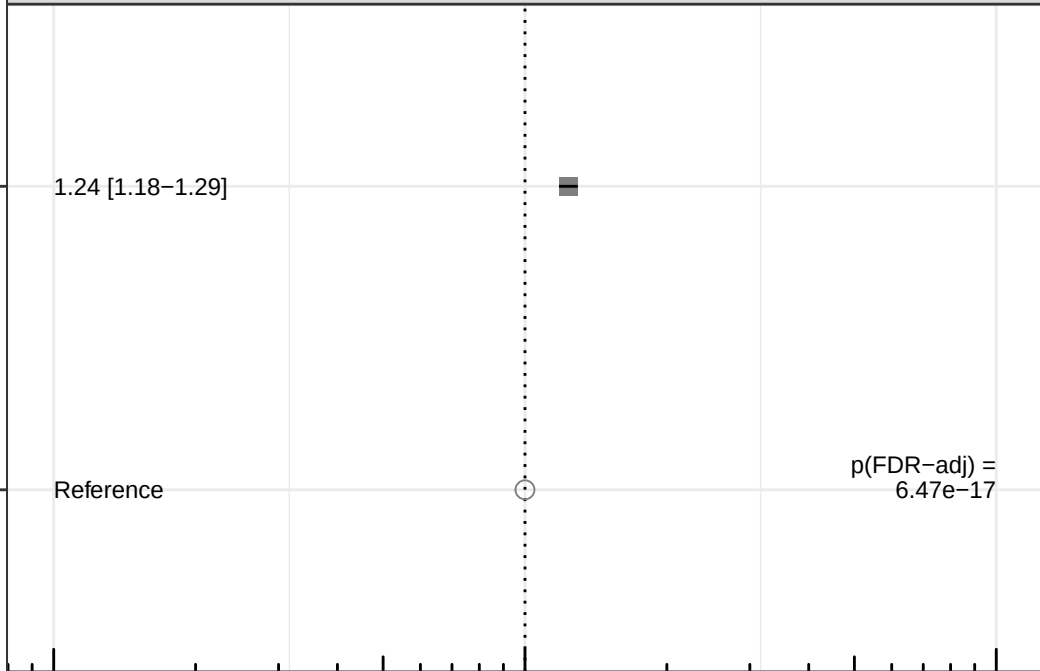
0.1

1.0

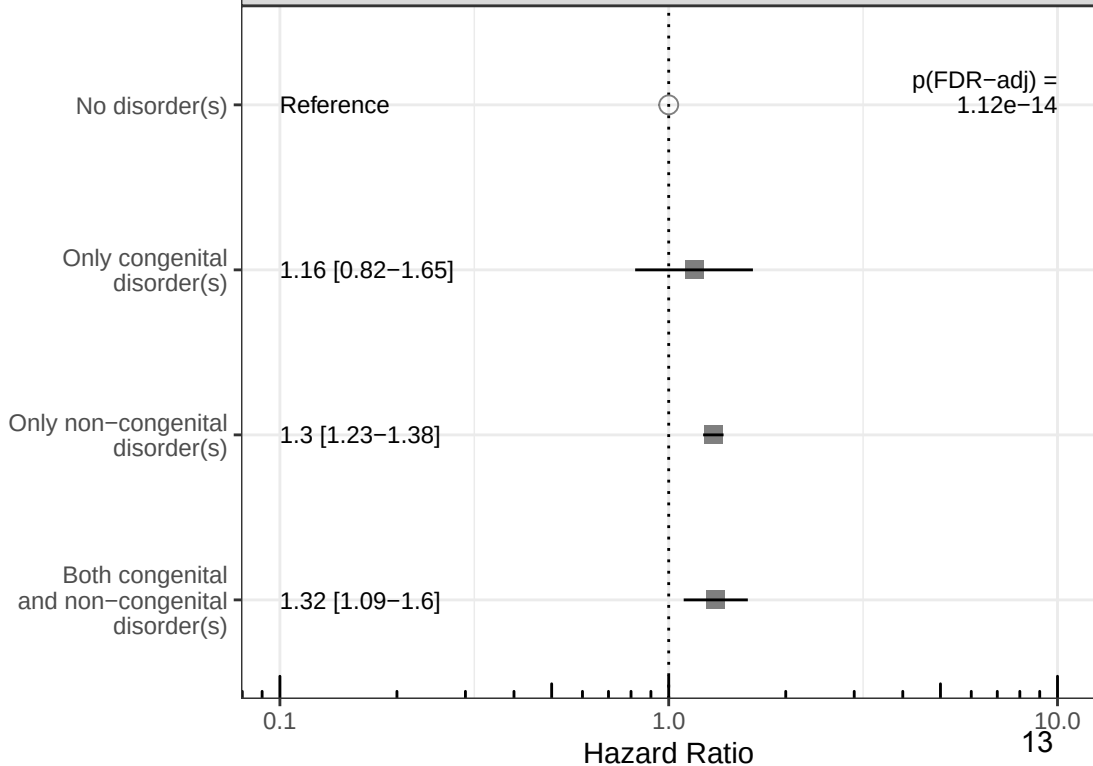
10.0

Hazard Ratio

12



hs\_health\_conditions\_cardiac  
Cardiac disorders



de\_recreational\_spaces  
Do you ever take your dog to parks or other open spaces meant for recreational activities?

Yes Reference  $p(\text{FDR-adj}) = 3.47\text{e-}14$

No 1.17 [1.13-1.22]

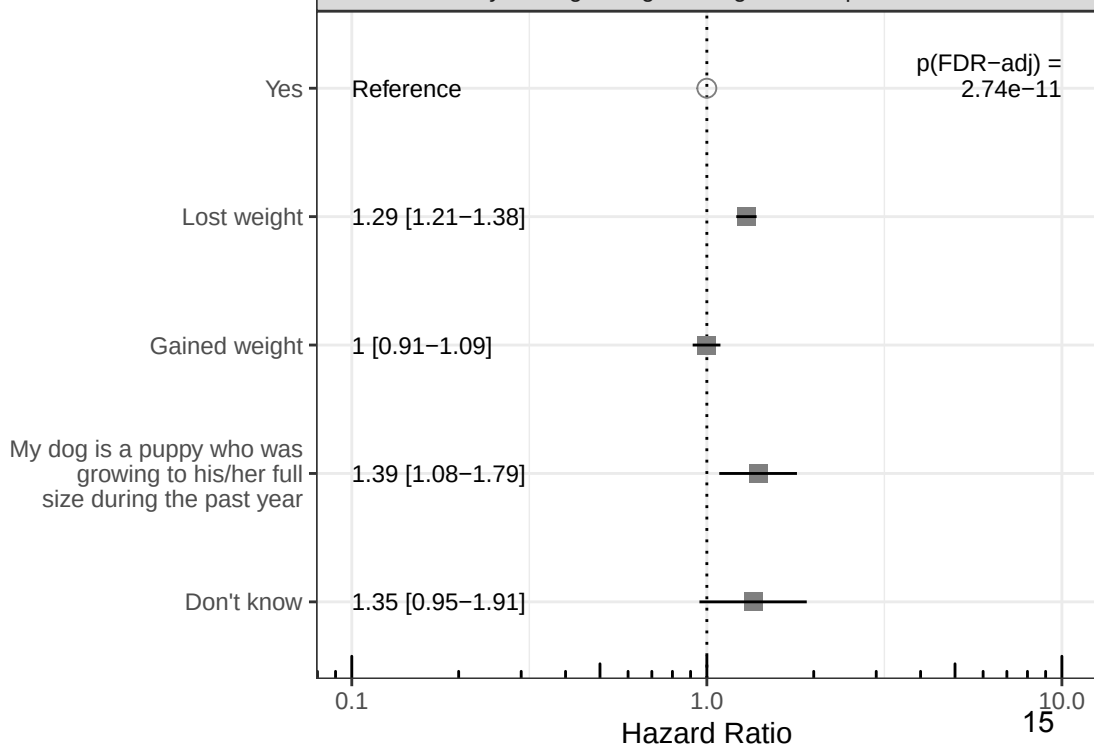
Hazard Ratio

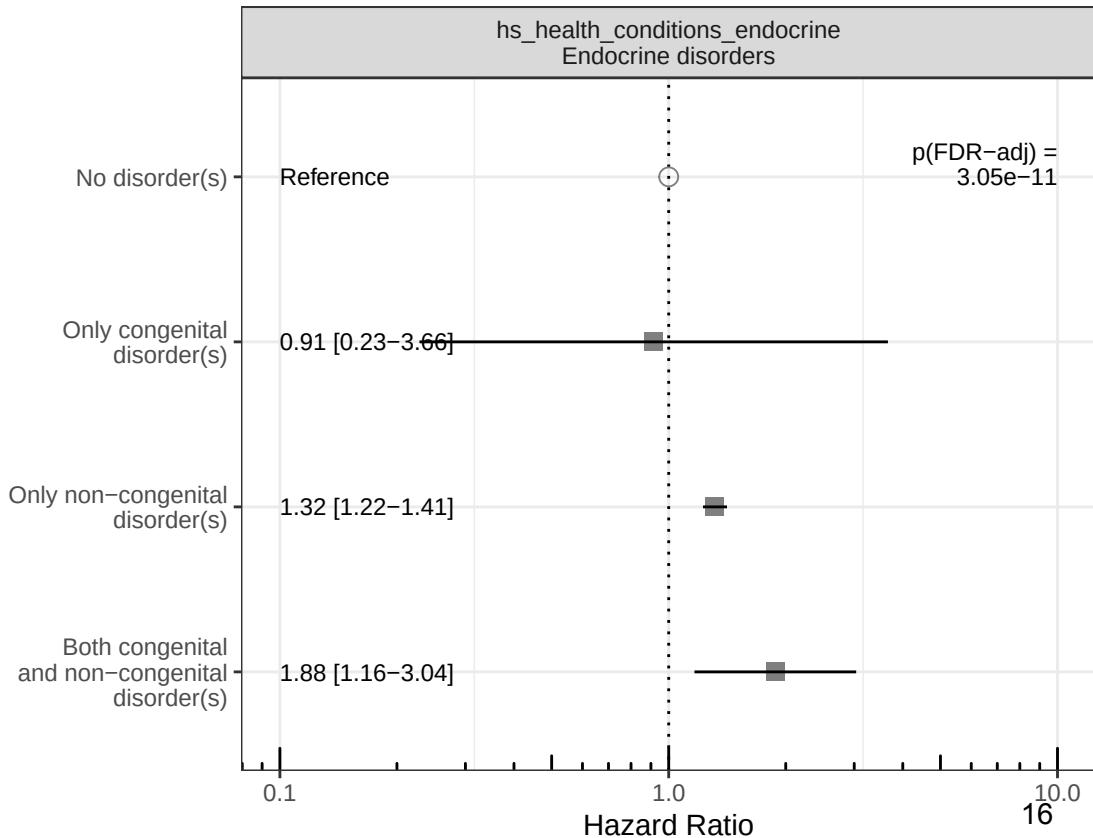
0.1

1.0

10.0  
14

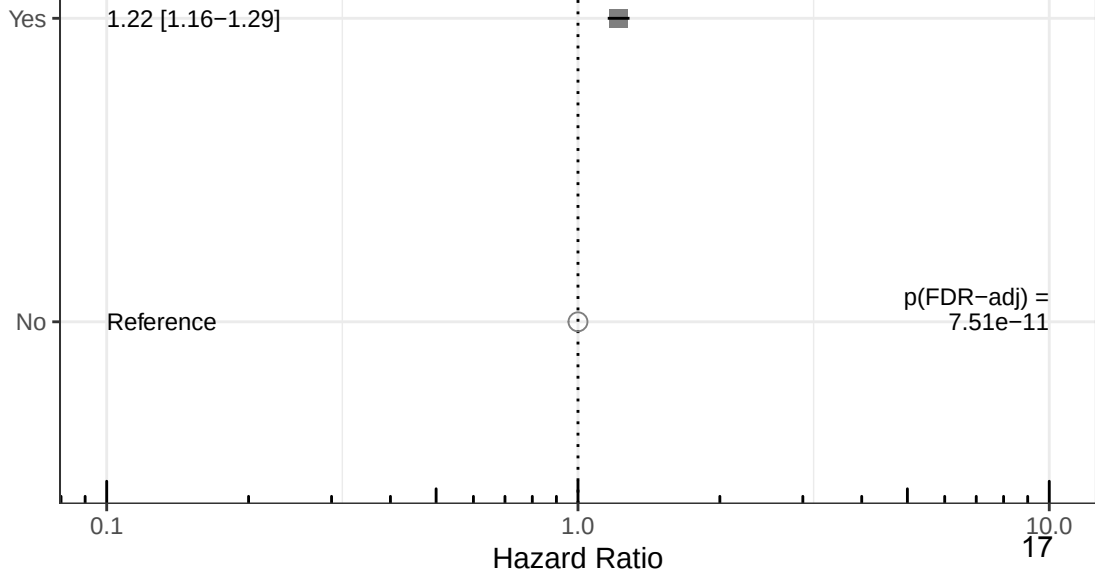
df\_weight\_change\_last\_year  
In the past 12 months, has your dog's weight stayed about the same?  
How has your dog's weight changed in the past 12 months?

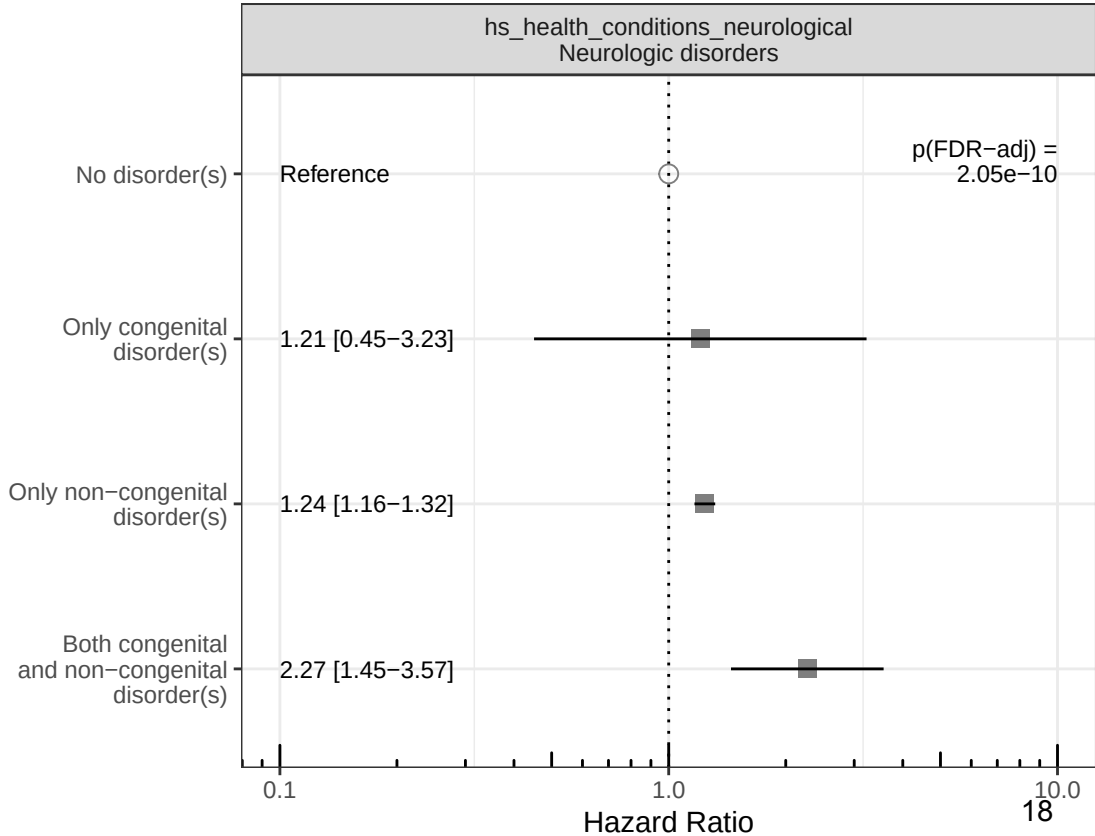




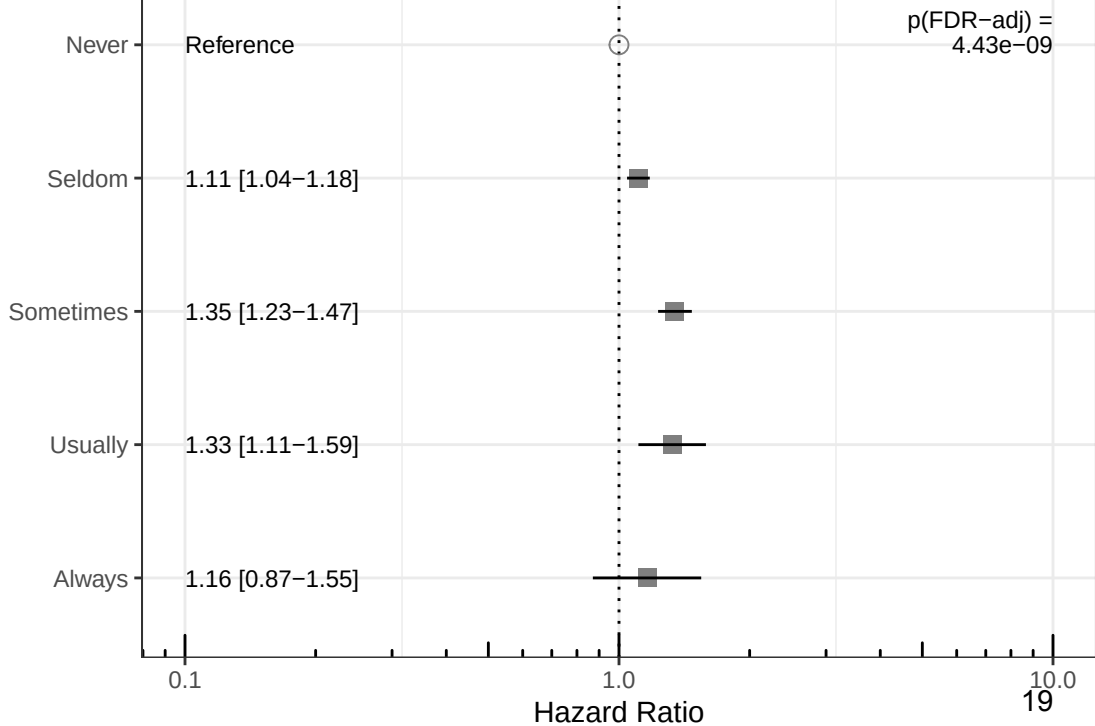


hs\_chronic\_condition\_recently\_changed\_or\_treated  
In the past three months, has your dog had any changes to the status  
of, or medications for, these conditions?

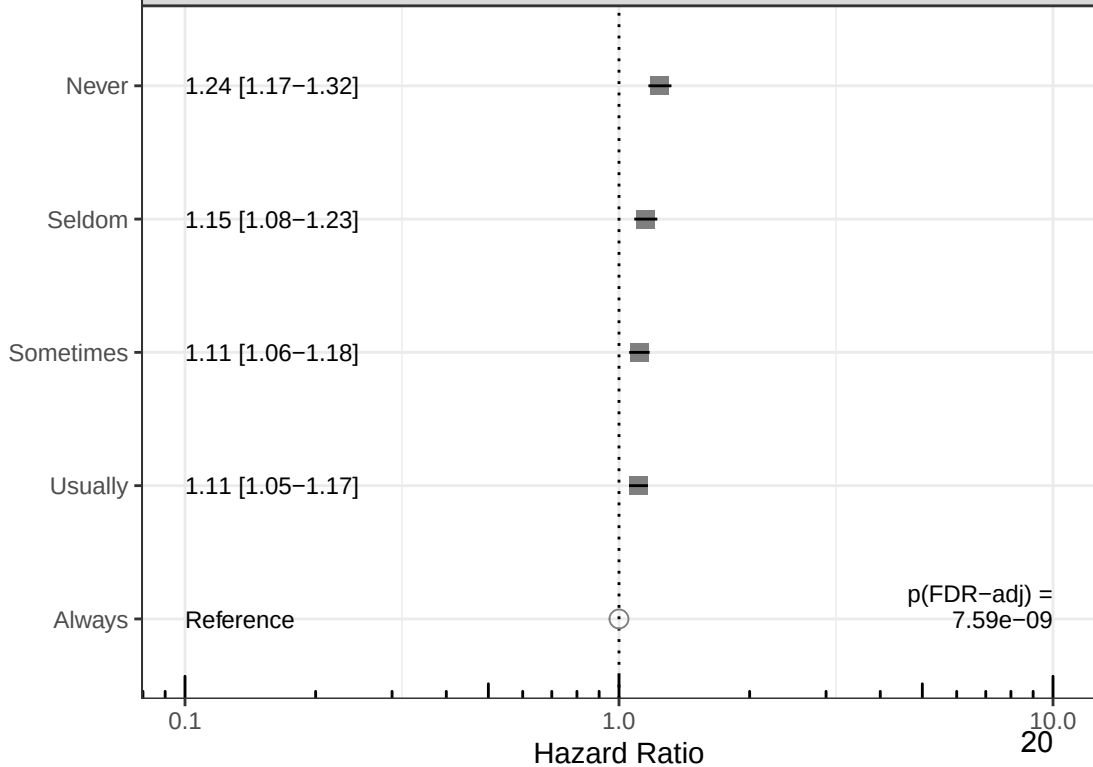




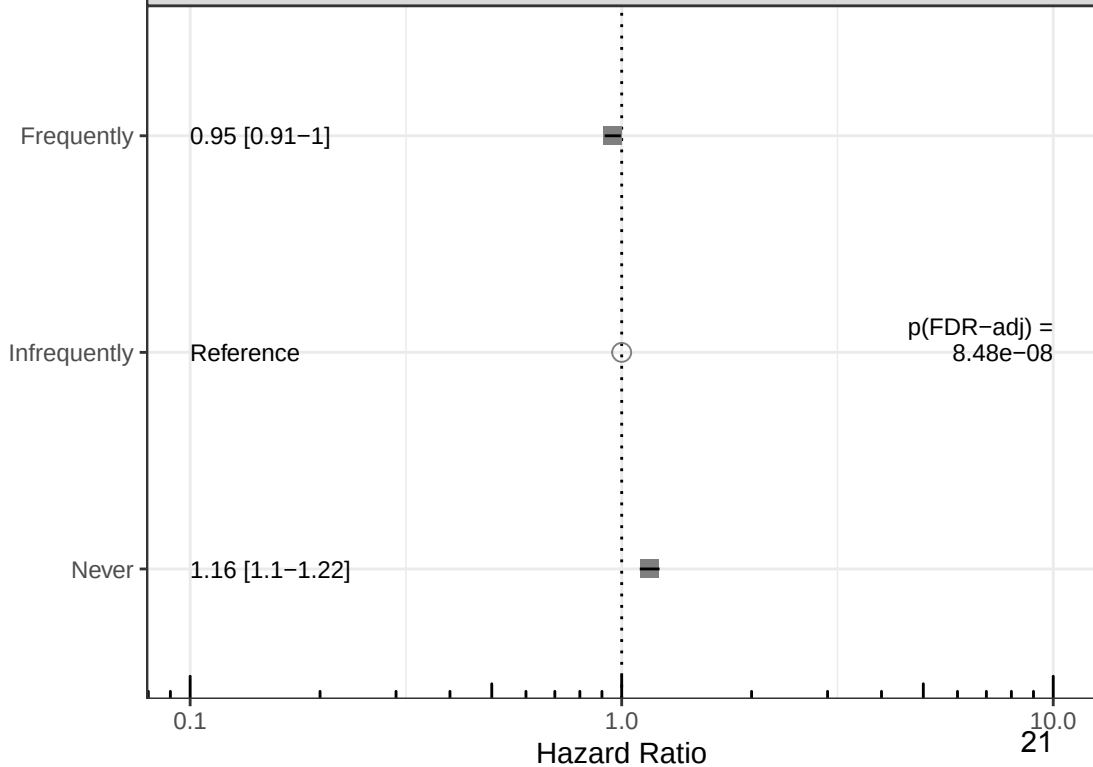
db\_defecates\_alone\_frequency  
Defecates when left alone at night or during the daytime:



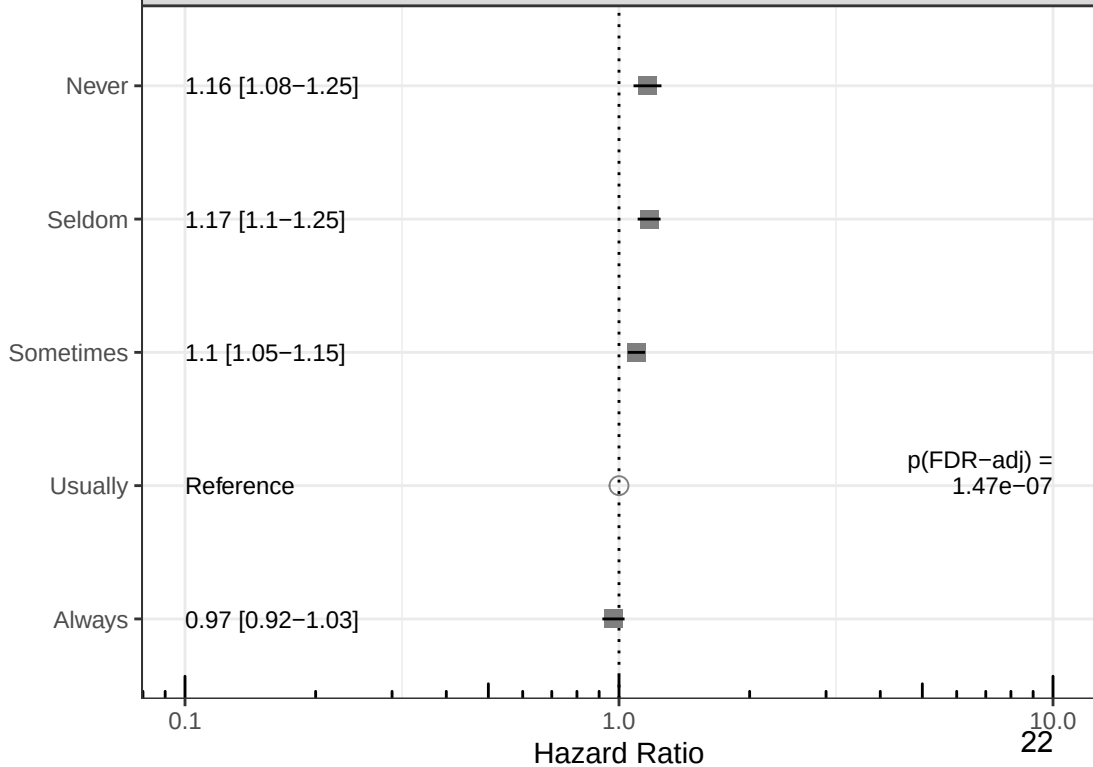
db\_chases\_squirrels\_frequency  
Chases or would chase squirrels, rabbits, etc., given the chance:



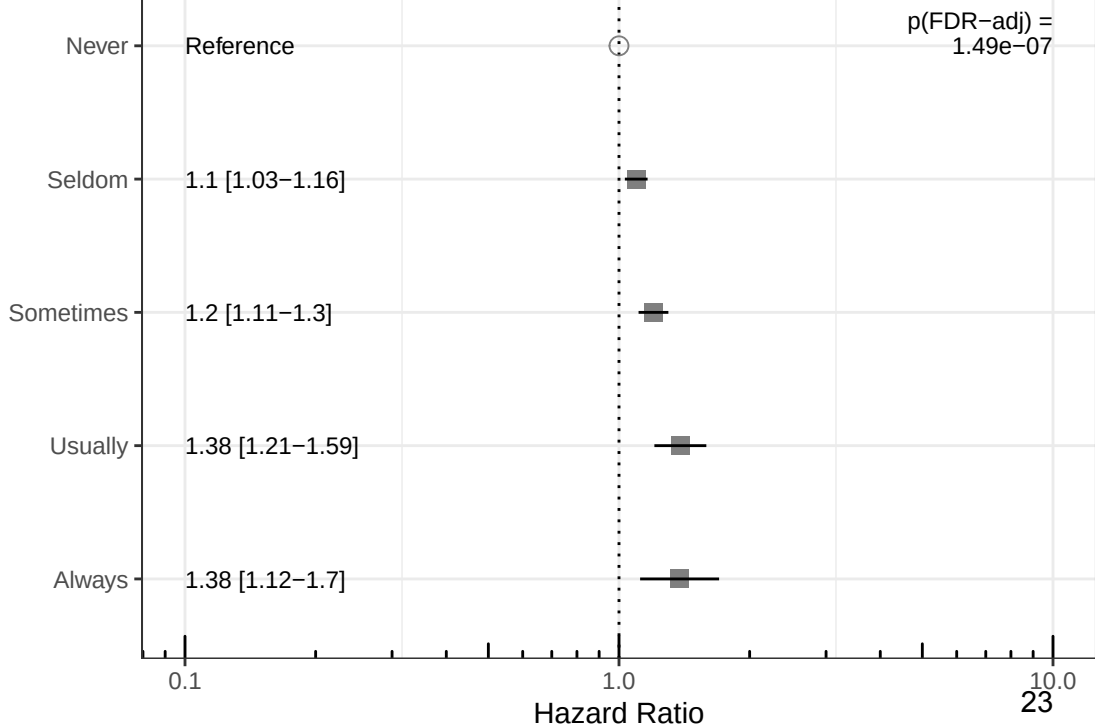
de\_eats\_grass\_frequency  
How often do you see your dog eating grass?

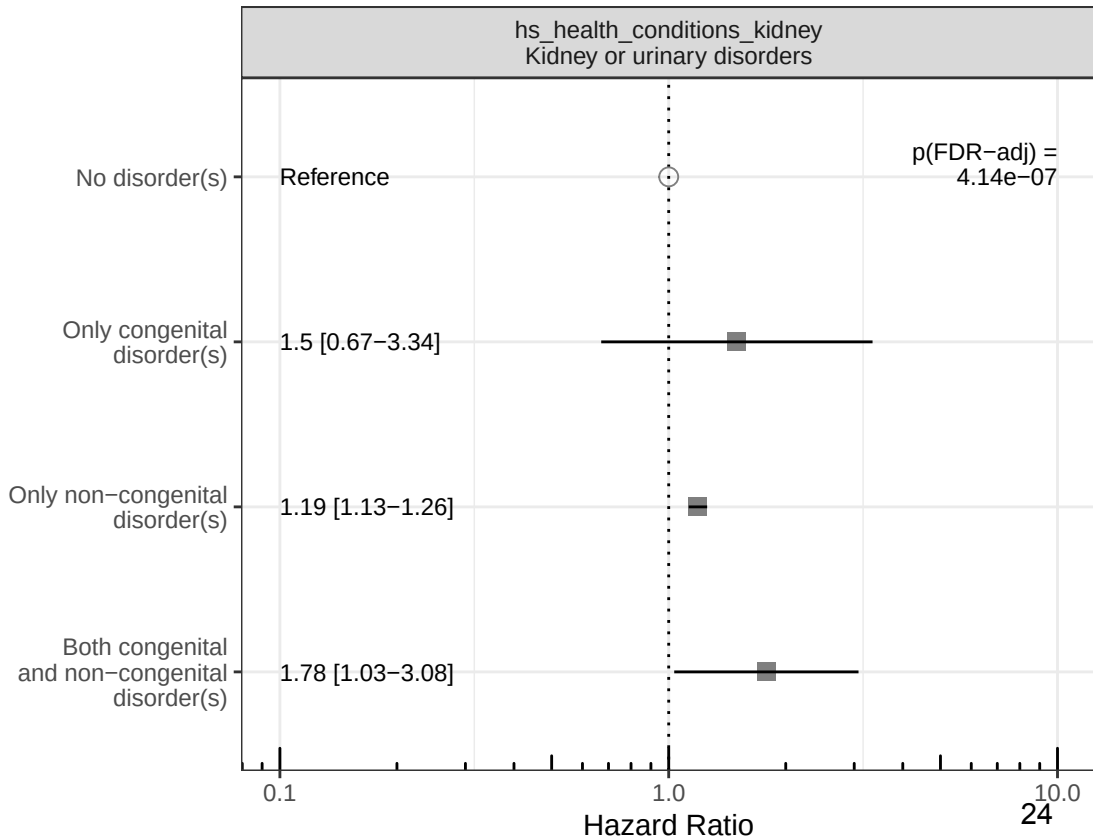


db\_training\_obeyes\_stay\_command\_frequency  
Obeys a "stay" command immediately:



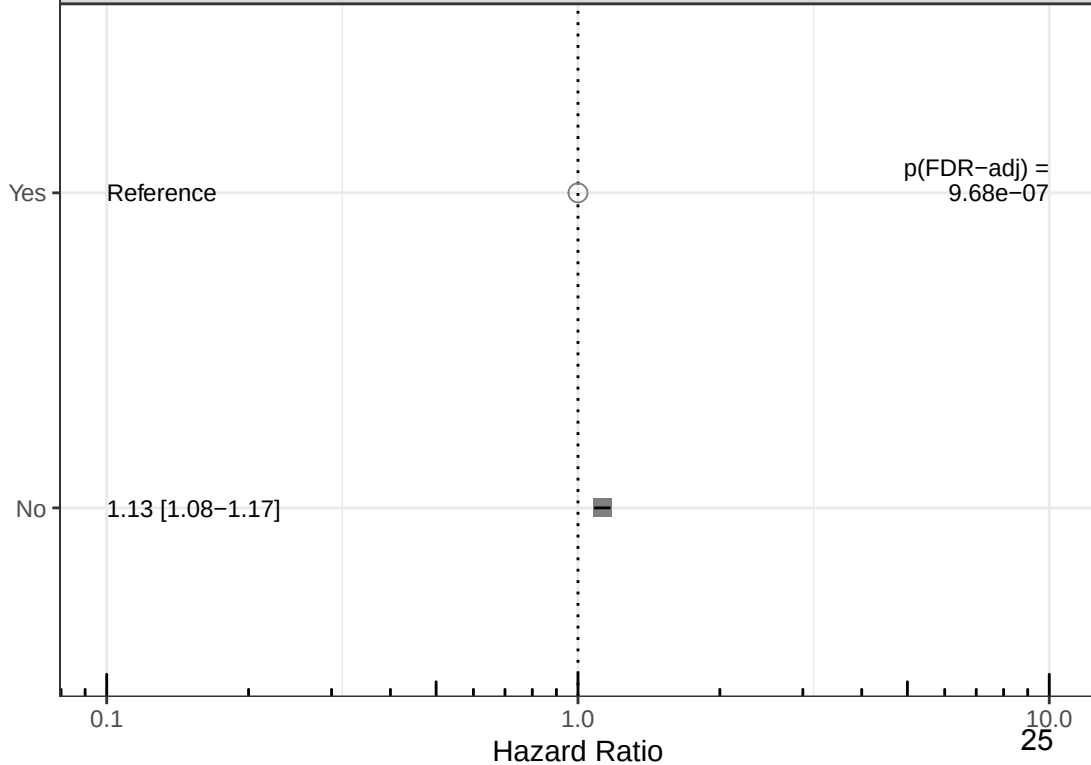
db\_urinates\_alone\_frequency  
Urinates when left alone at night or during the daytime:







pa\_on\_leash\_walk\_reasons\_exercise\_for\_owner  
What is the reason for your activities? Exercise for you



de\_routine\_toys  
Does your dog regularly lick, chew, or play with toys?

Yes Reference  $p(\text{FDR-adj}) = 1.04\text{e-}06$

No 1.12 [1.08–1.17]

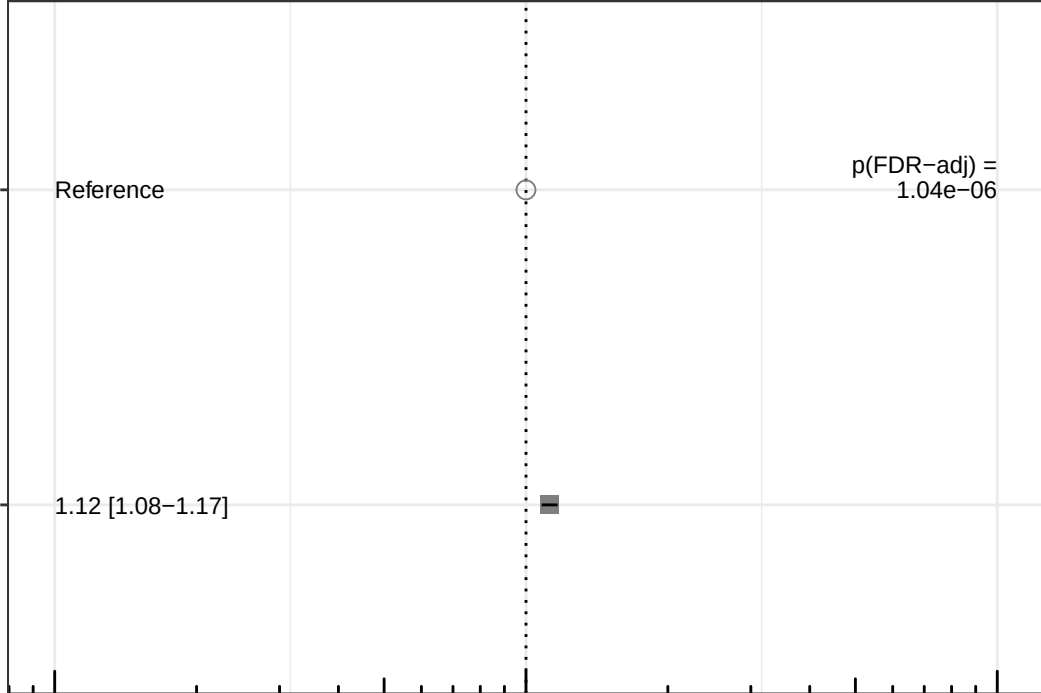
0.1

1.0

10.0

Hazard Ratio

26



de\_stairs\_in\_home  
Is there a staircase with more than 8 stairs that your dog uses regularly?

Yes Reference  $p(\text{FDR-adj}) = 1.44\text{e-}06$

No 1.11 [1.07-1.16]

0.1

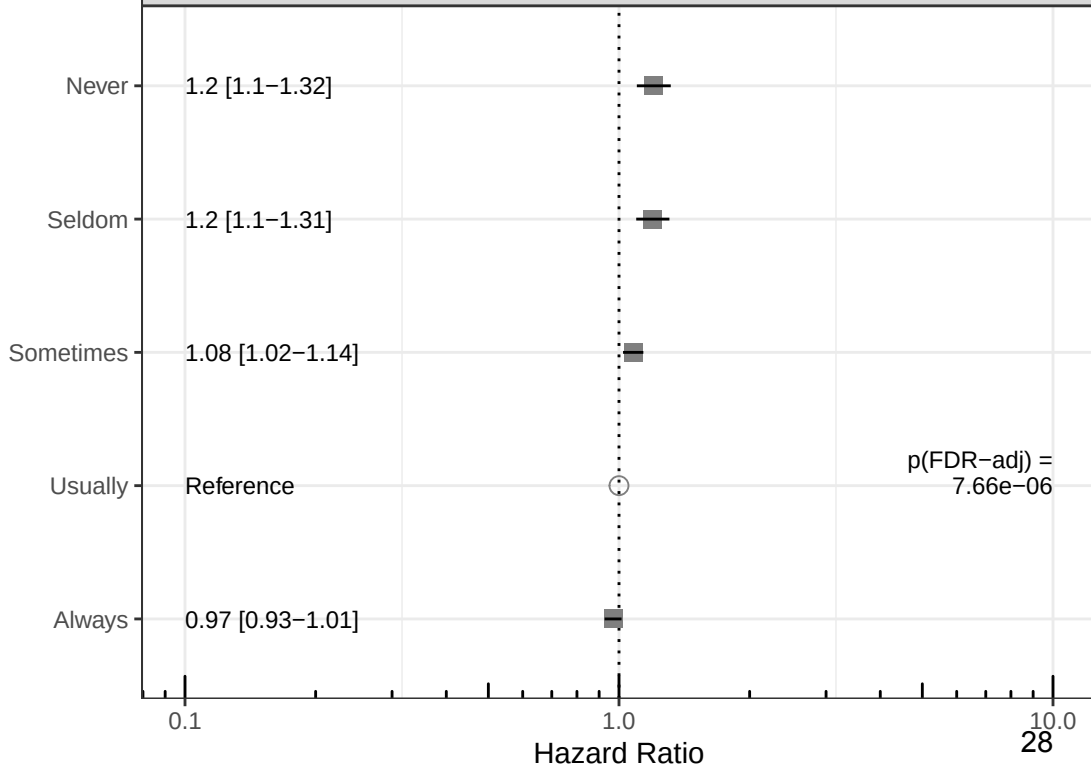
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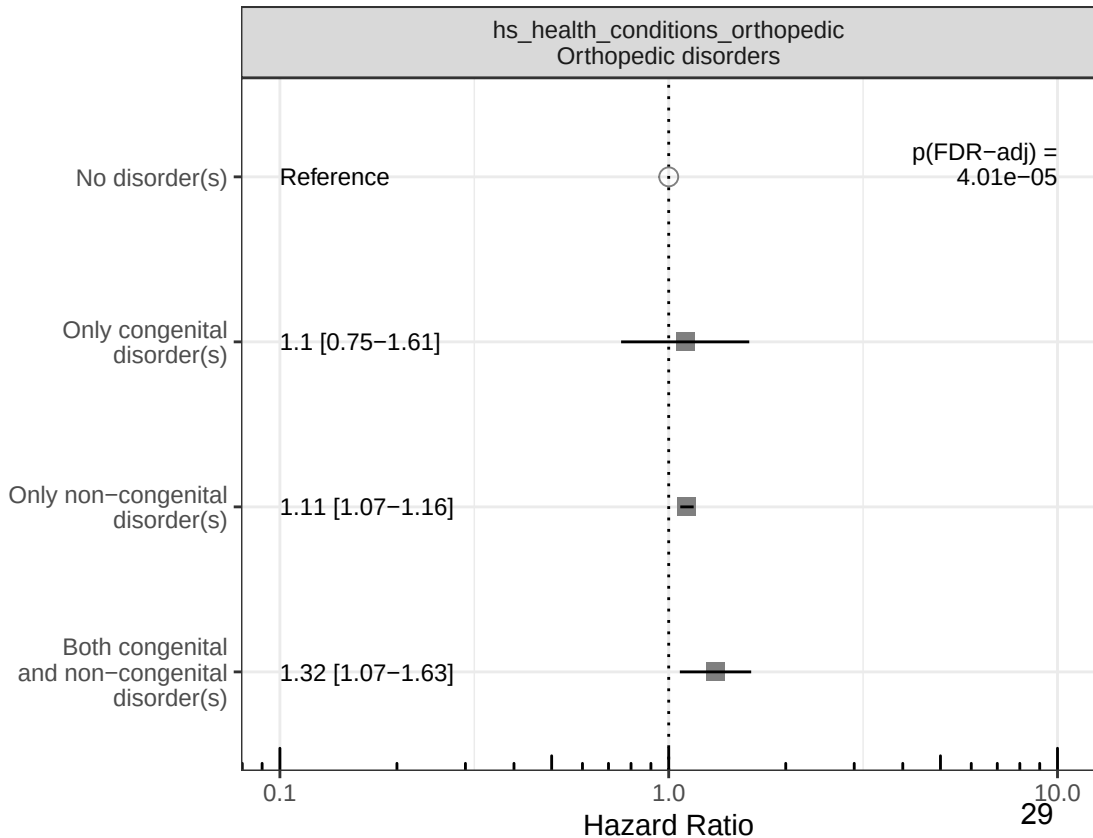
10.0

Hazard Ratio

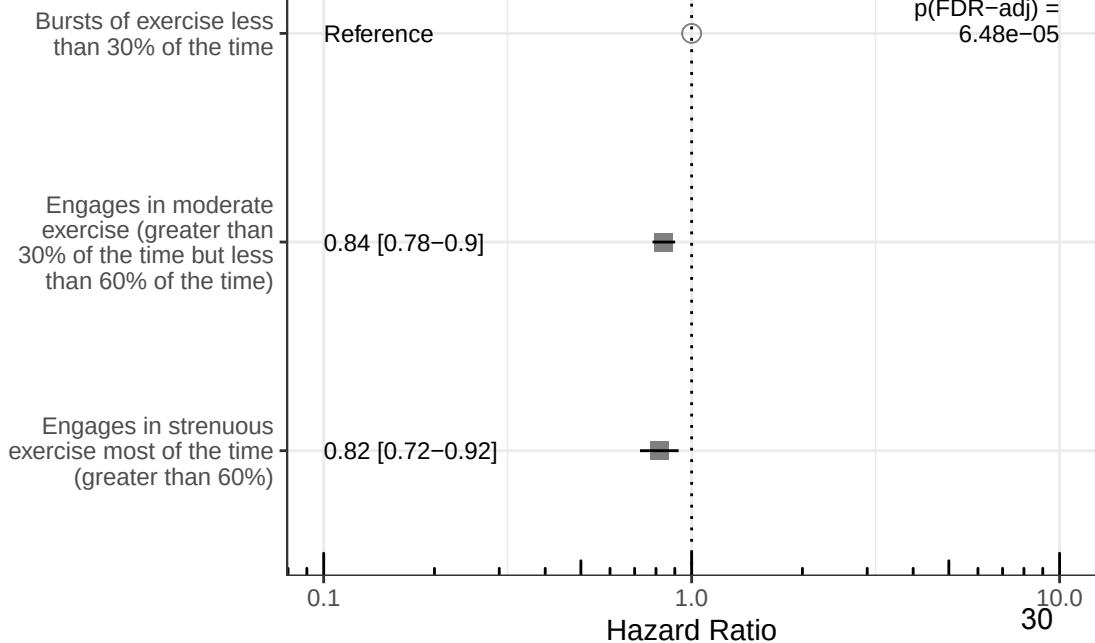
27

db\_training\_obeyes\_sit\_command\_frequency  
Obeys a "sit" command immediately:

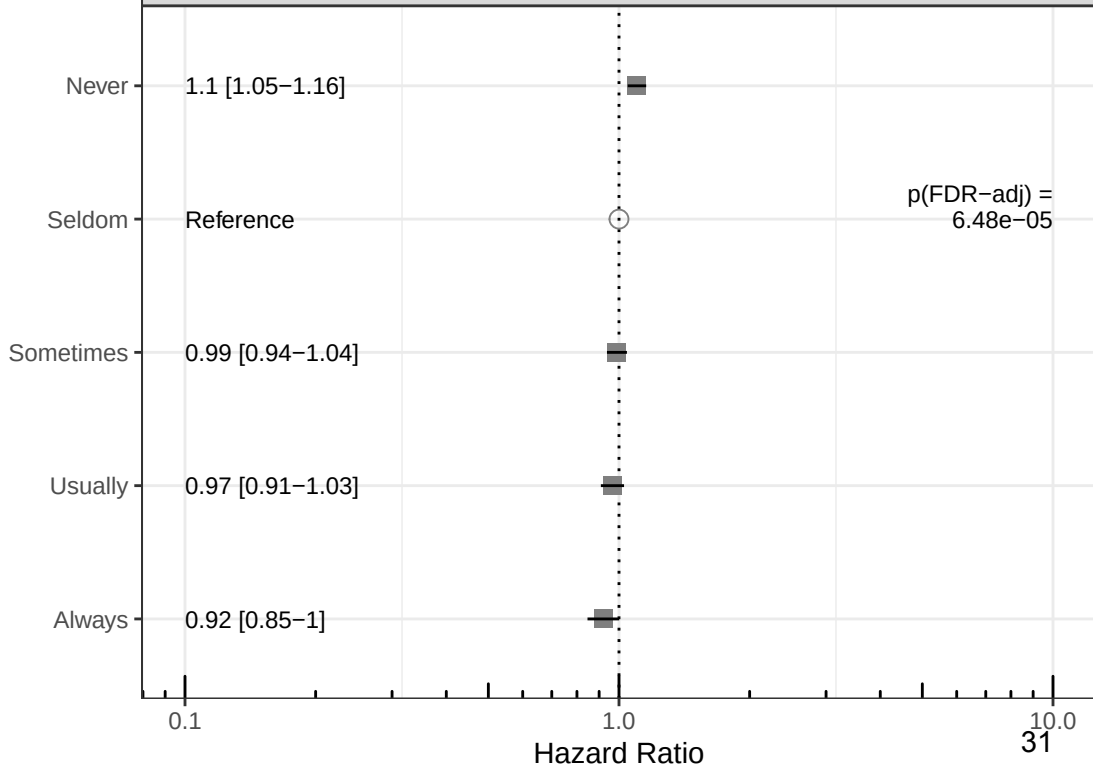




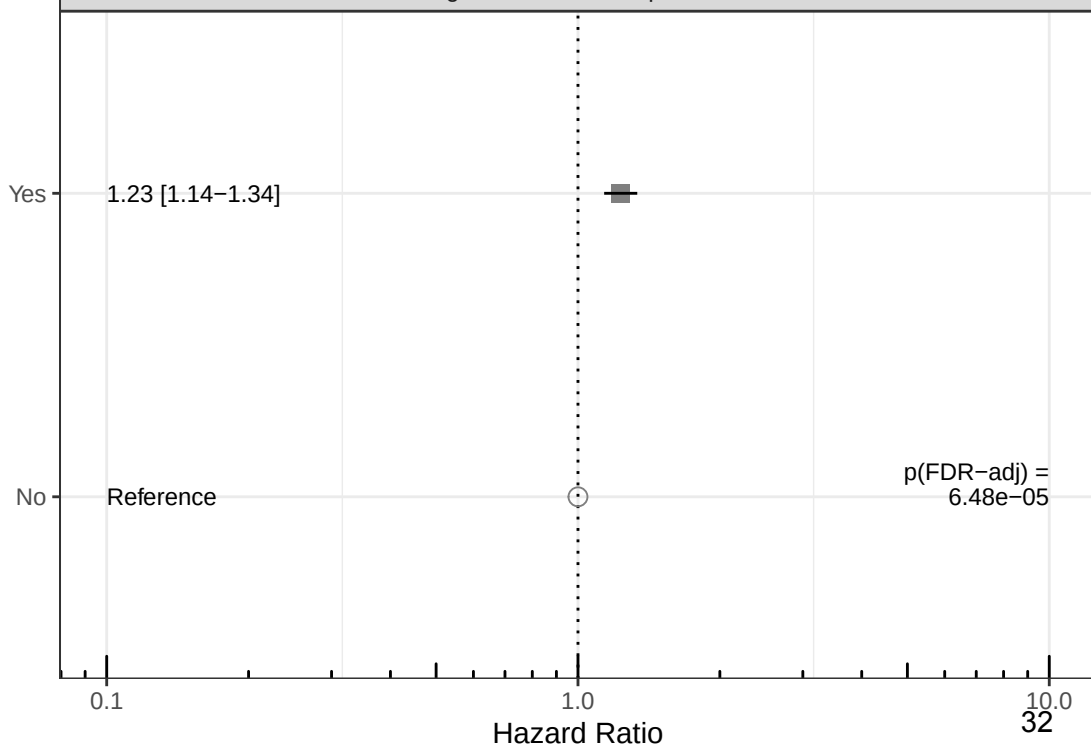
pa\_other\_aerobic\_activity\_avg\_intensity  
During these periods of other aerobic activity, which of the following  
best describes your dog's activity level?



db\_barks\_frequency  
Barks persistently when alarmed or excited:

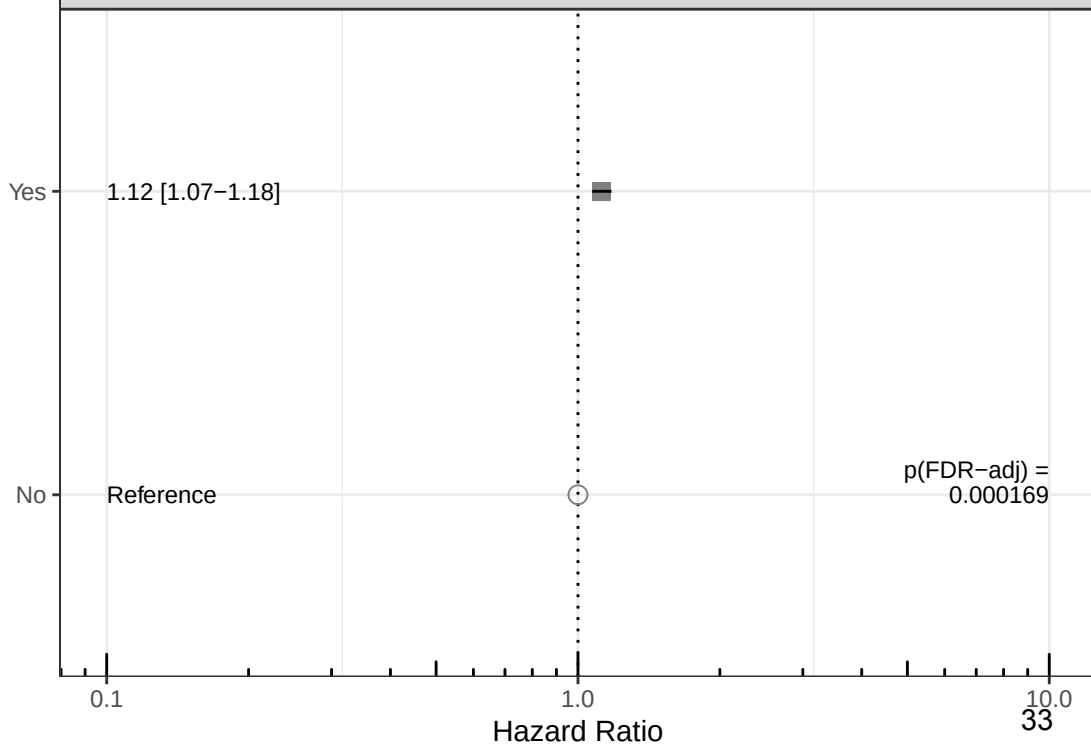


df\_primary\_diet\_component\_change\_health\_condition\_specific  
What were the reasons for the diet change? Health diagnosis prompted  
change to a condition-specific diet

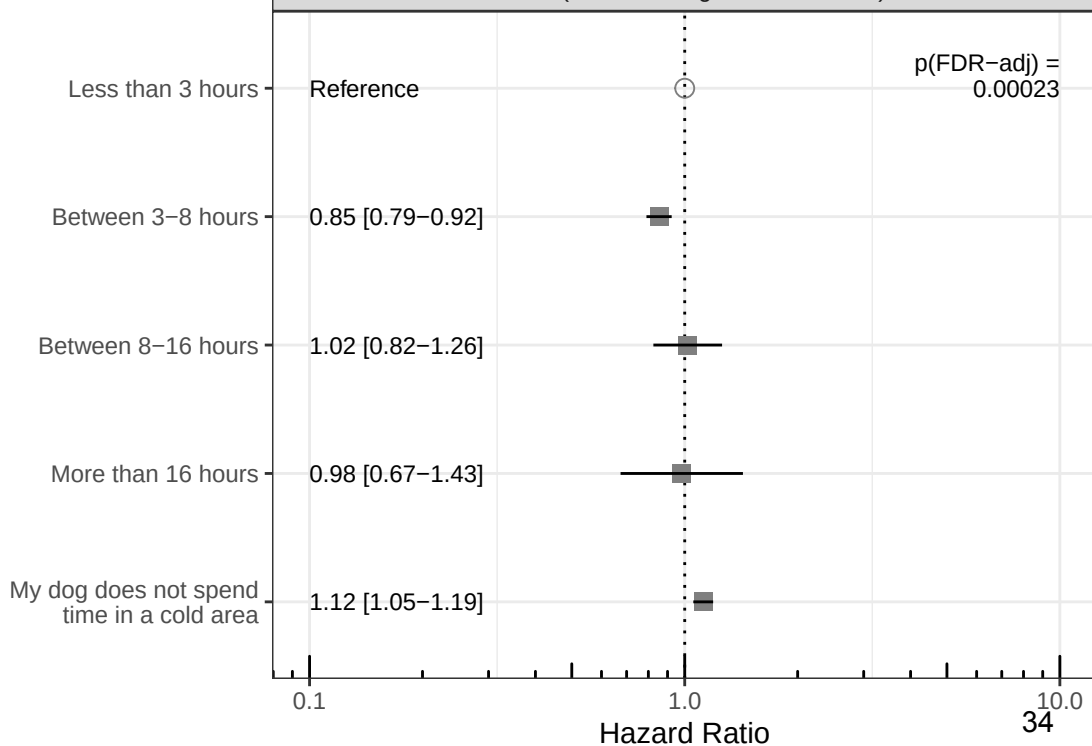




mp\_recent\_non\_prescription\_meds\_upset\_stomach  
Given in past year: Medications for stomach or GI upset (such as  
Pepcid)



pa\_cold\_weather\_daily\_hours\_outside  
How much time per day does your dog spend outdoors on average in cold weather (below 40 degrees Fahrenheit)?



df\_primary\_diet\_component\_change\_recent

Have you made any changes to the primary component of your dog's diet  
in the past 12 months?

Yes 1.11 [1.06–1.16]

No Reference

p(FDR-adj) =  
0.000234

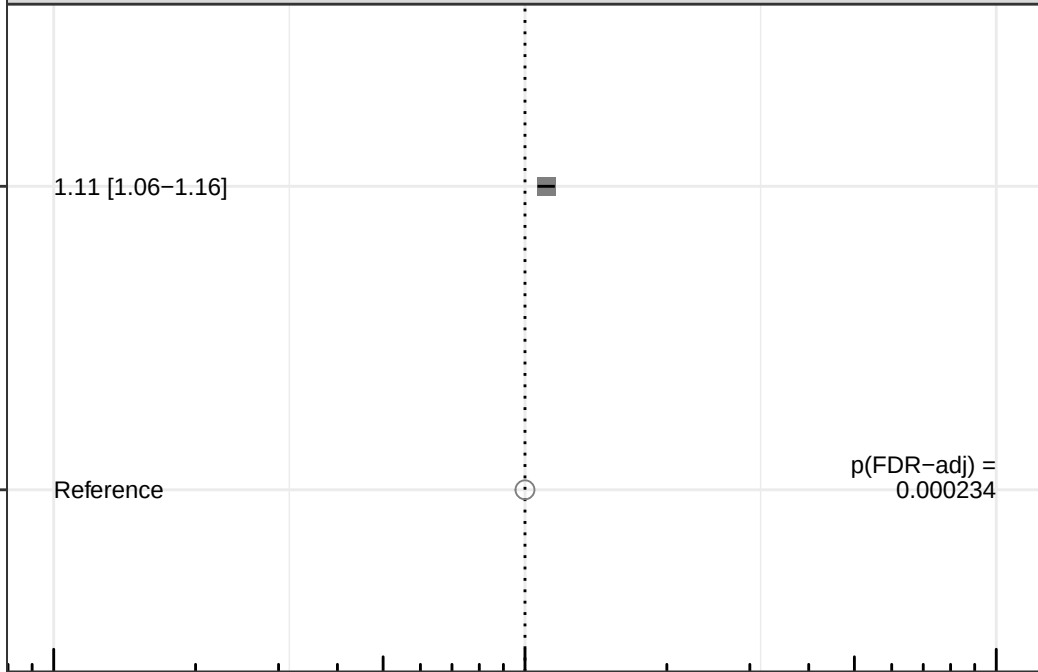
0.1

1.0

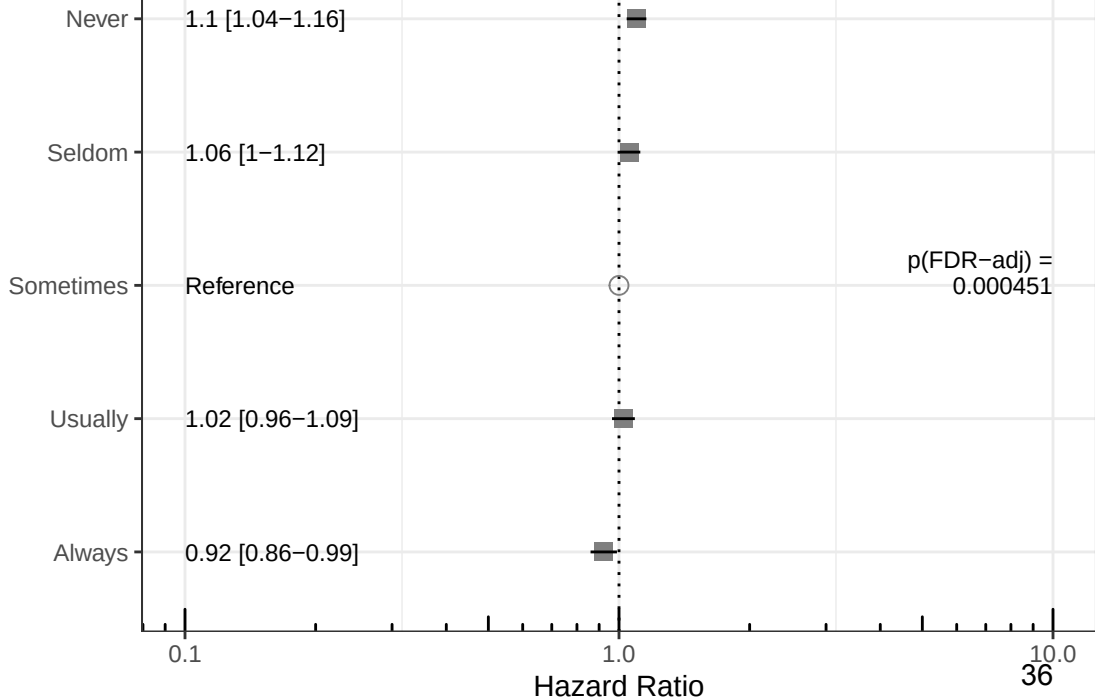
10.0

Hazard Ratio

35



db\_chases\_birds\_frequency  
Chases or would chase birds, given the chance:



de\_drinks\_outdoor\_water  
Does your dog drink from outdoor water sources (such as lake, river,  
stream, pond, puddles, etc.)?

Yes 0.91 [0.87–0.95]

No Reference

p(FDR-adj) =  
0.000575

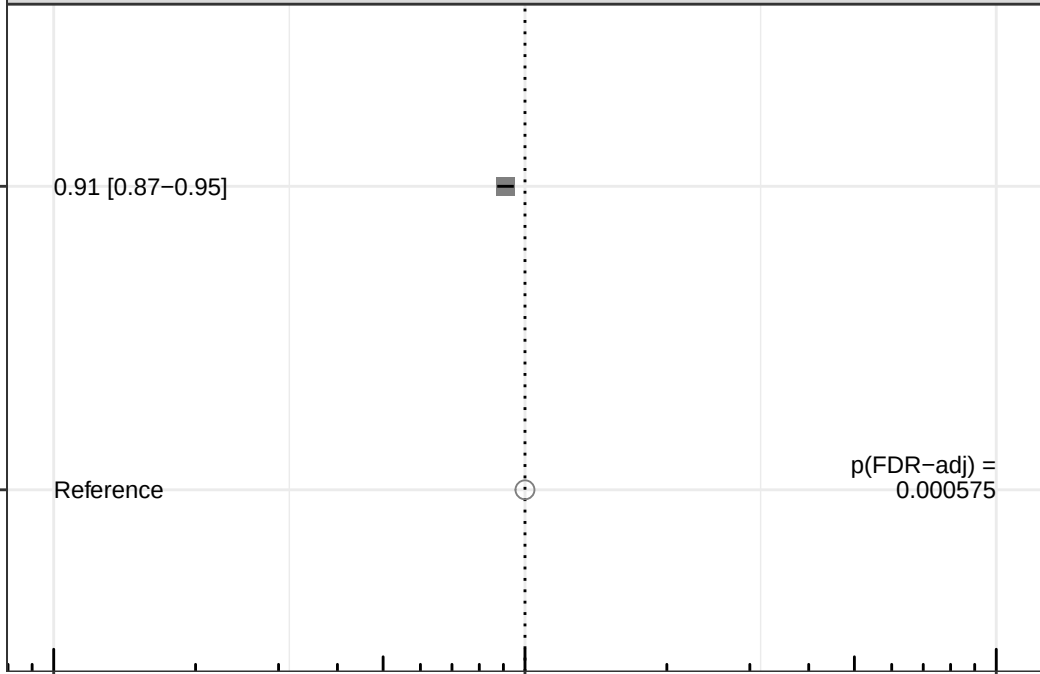
Hazard Ratio

0.1

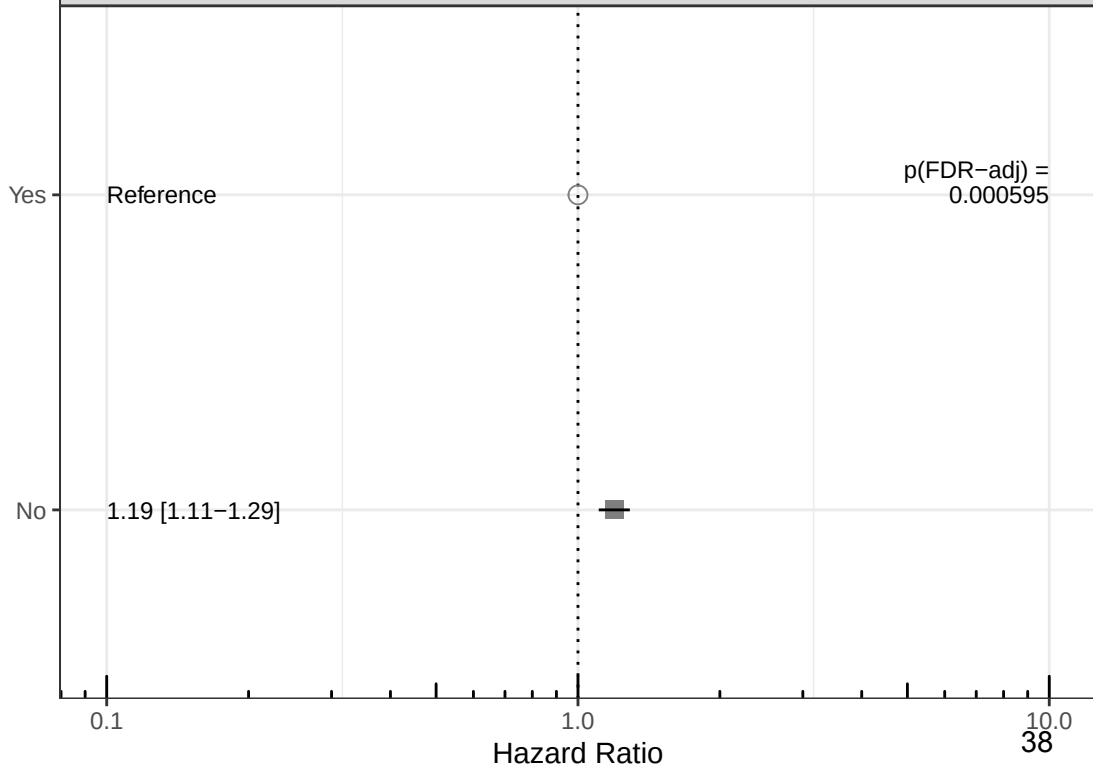
1.0

10.0

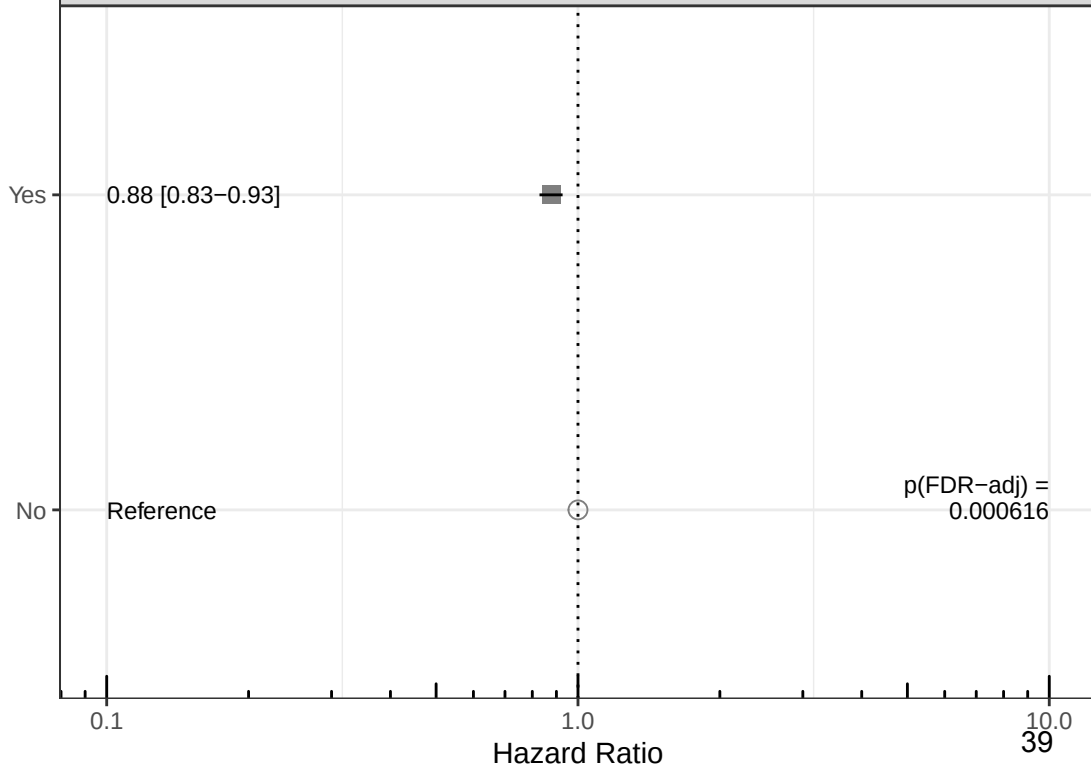
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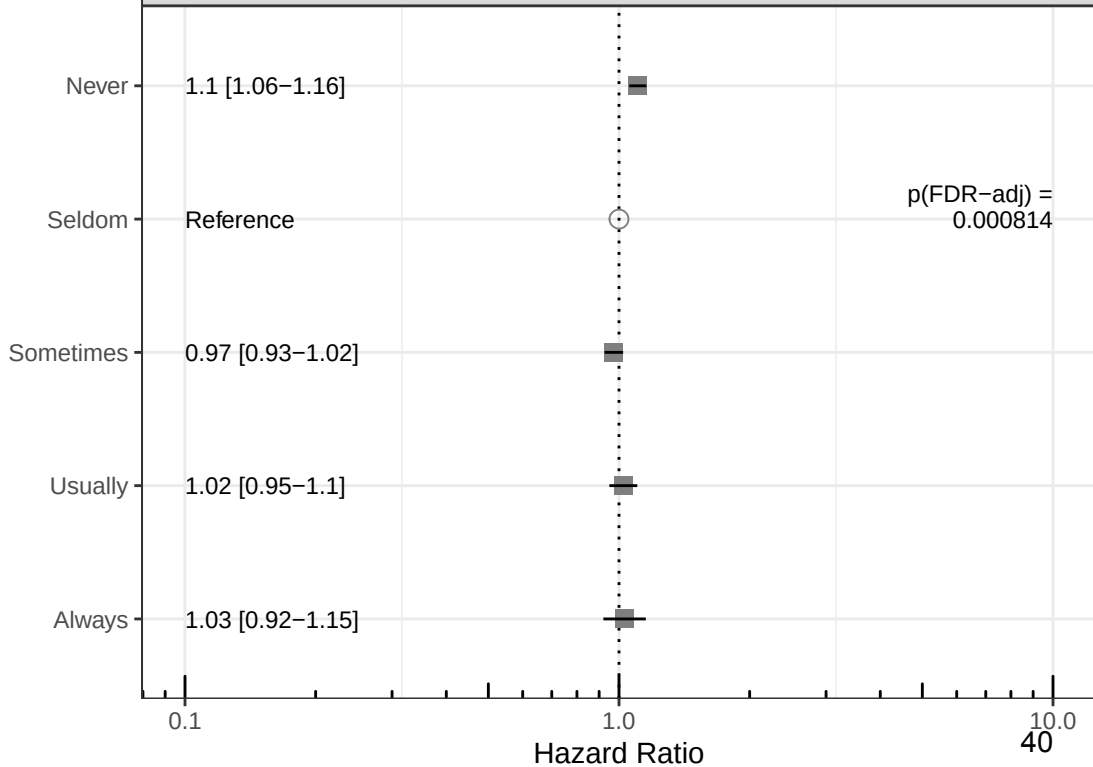
pa\_off\_leash\_walk\_reasons\_activity\_and\_enjoyment  
What is the reason for your activities? General activity and enjoyment



pa\_moderate\_weather\_outdoor\_sand  
Sand:

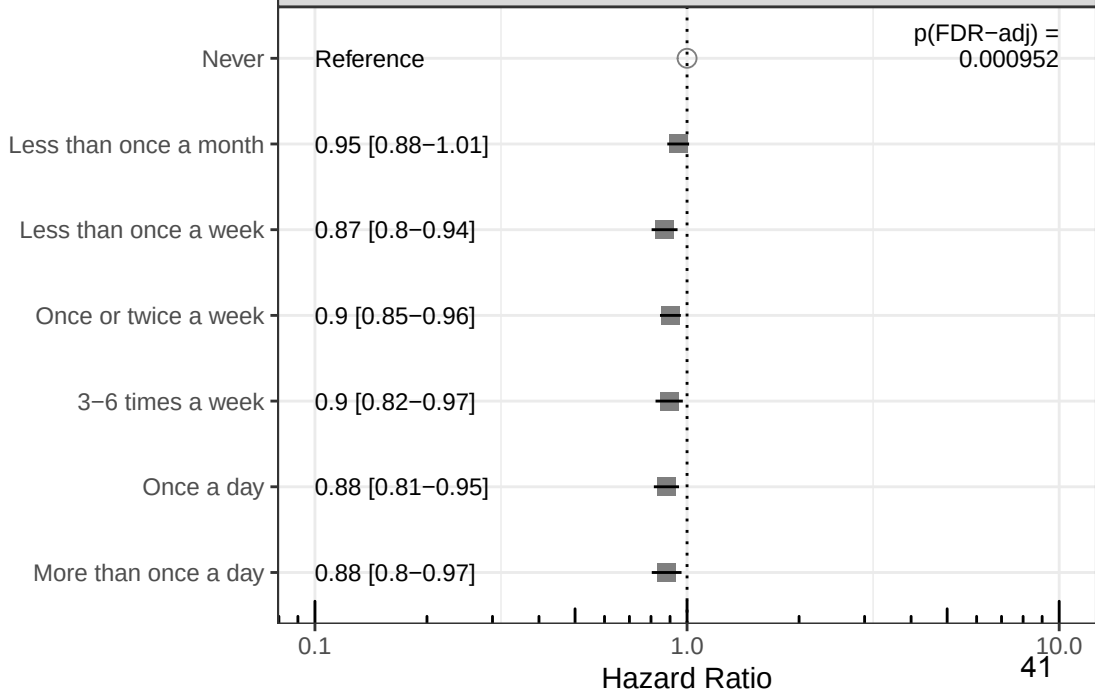


db\_pulls\_leash\_frequency  
Pulls excessively hard when on the leash:

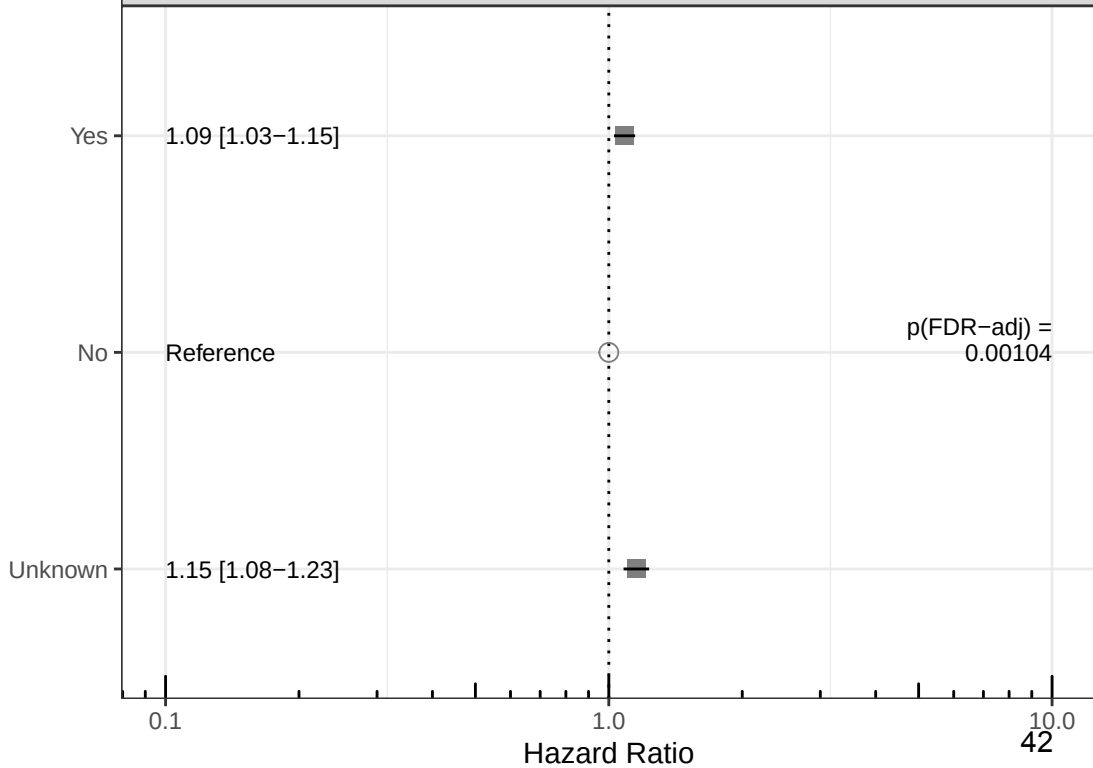




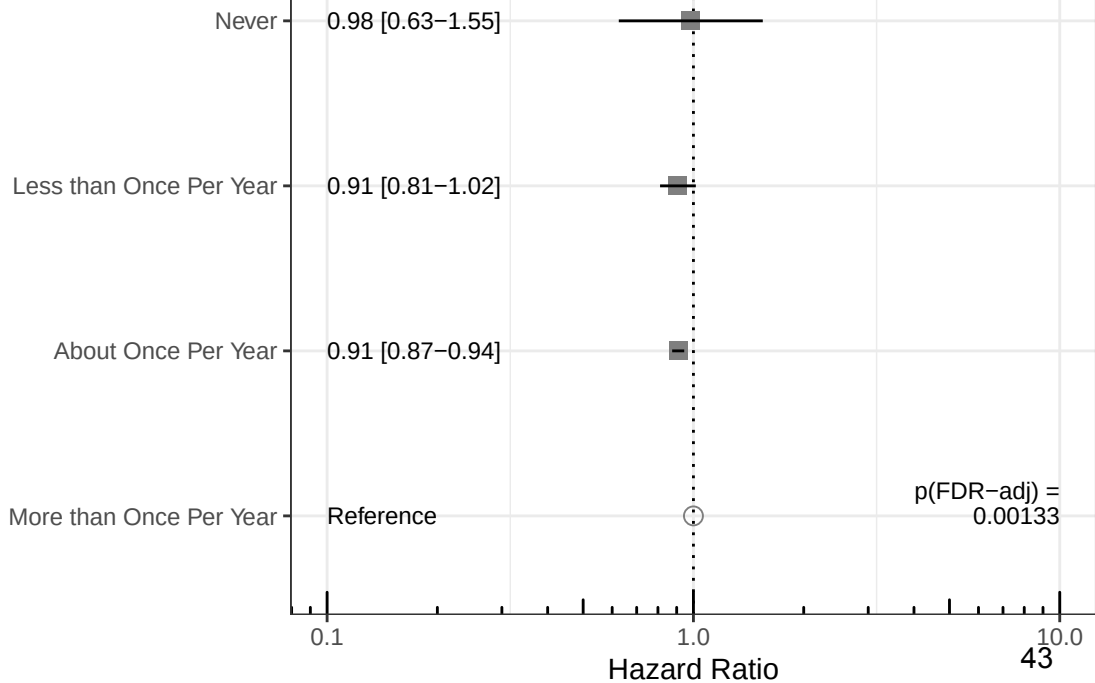
pa\_other\_aerobic\_activity\_frequency  
 Other than the on-leash and off-leash activities already described above, what is the average frequency that your dog gets other aerobic (elevated heart rate) activity? These might include working, mushing, or herding.



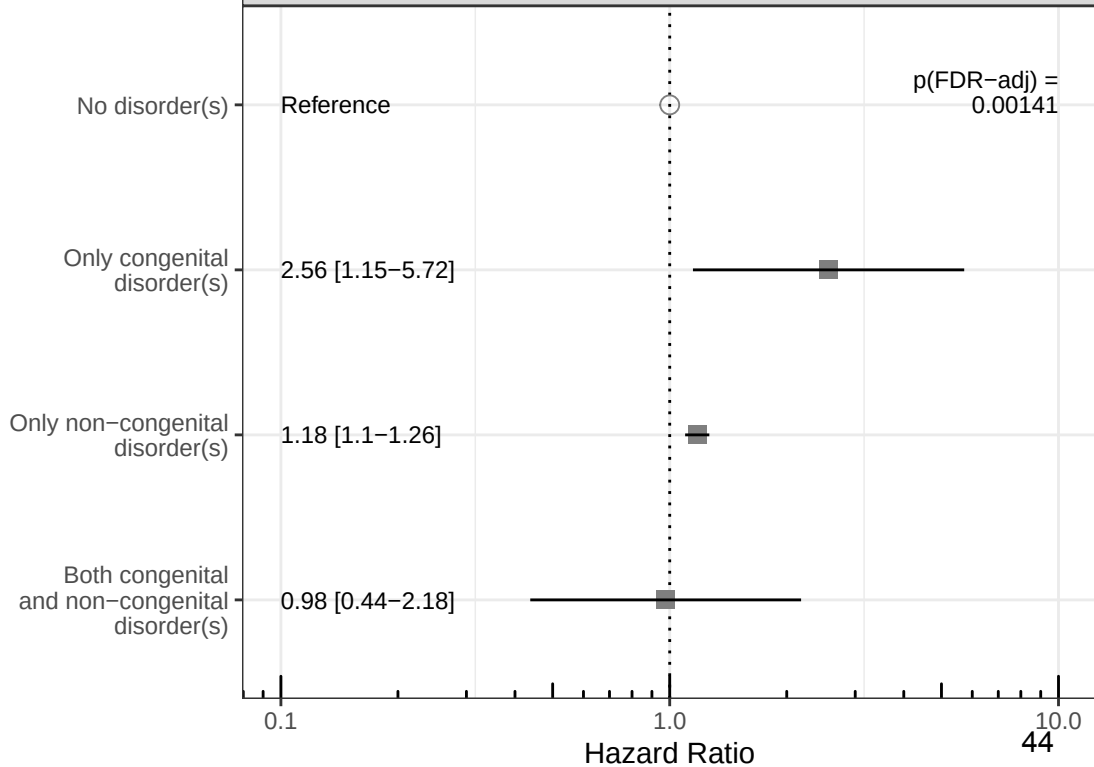
df\_ever\_underweight  
Has your dog ever been considered underweight?



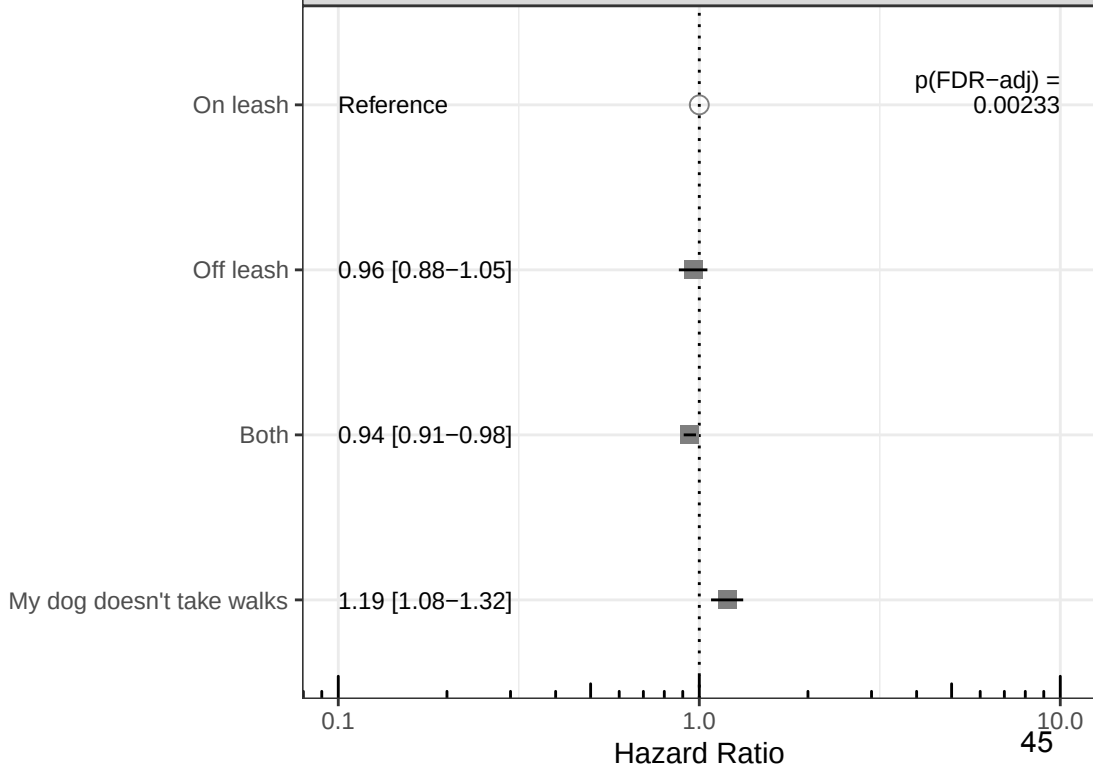
hs\_vet\_frequency  
How often is your dog seen by a veterinarian?



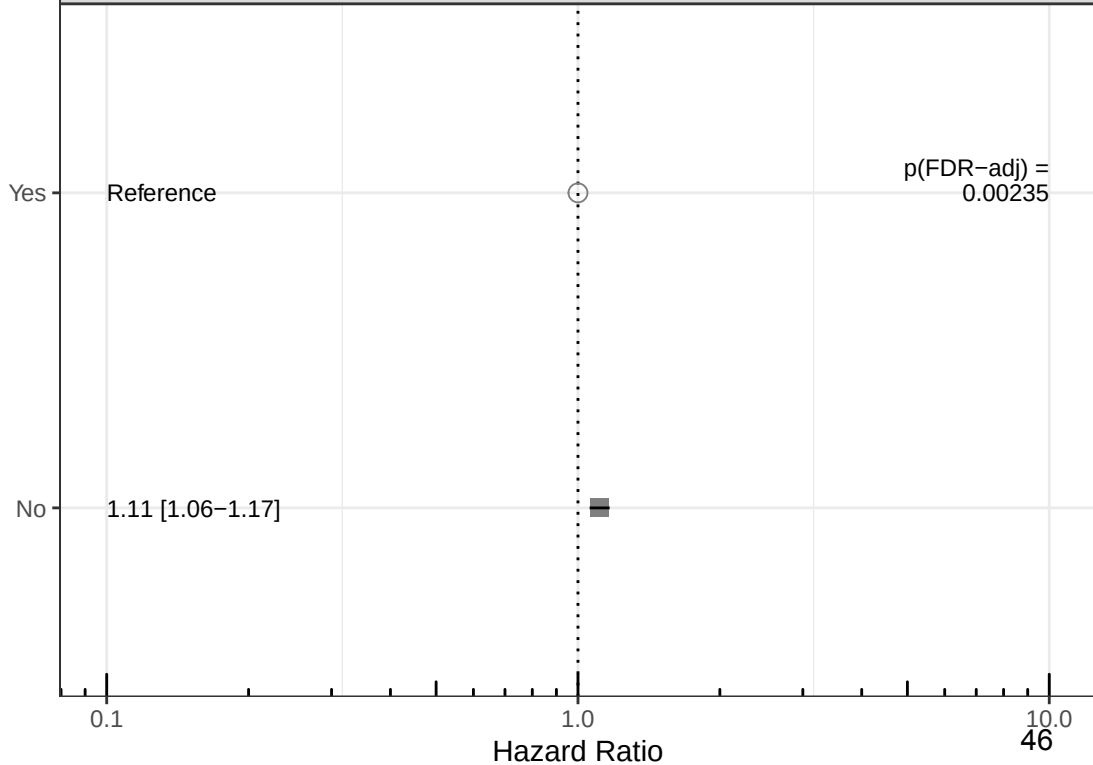
hs\_health\_conditions\_liver  
Liver or pancreas disorders



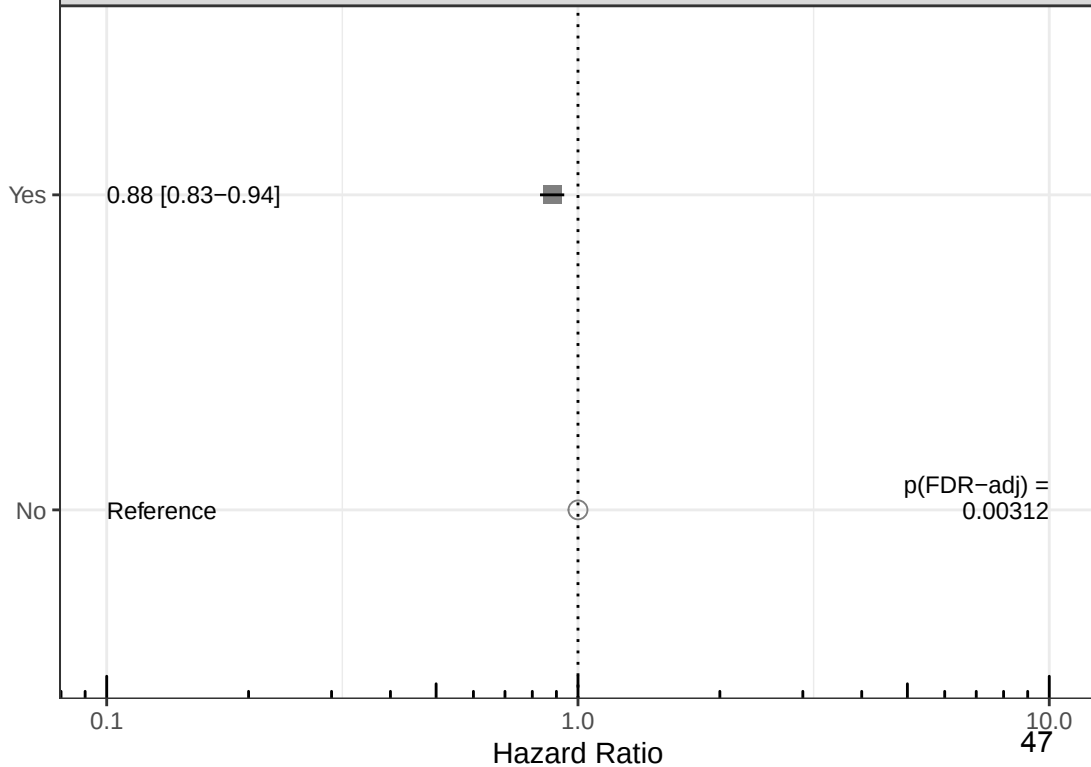
pa\_on\_leash\_off\_leash\_walk  
Over the past year, if your dog leaves the house with you, they are:



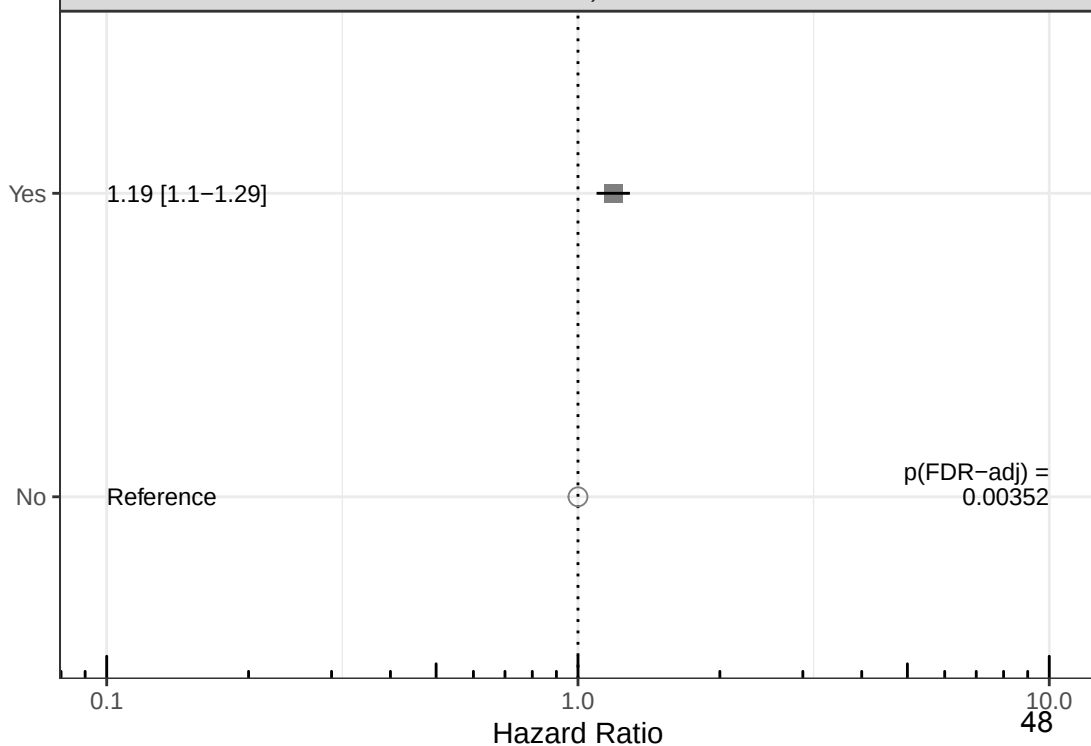
pa\_on\_leash\_walk\_reasons\_exercise\_for\_dog  
What is the reason for your activities? Exercise for the dog



pa\_off\_leash\_walk\_reasons\_exercise\_for\_owner  
What is the reason for your activities? Exercise for you

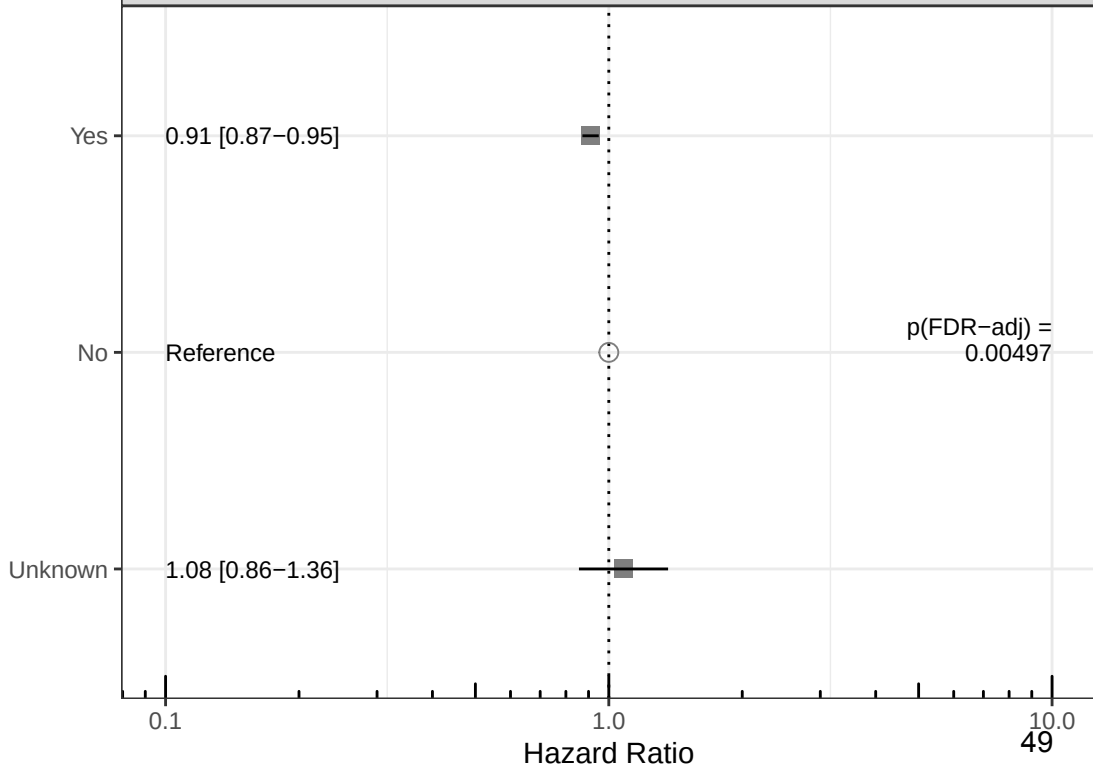


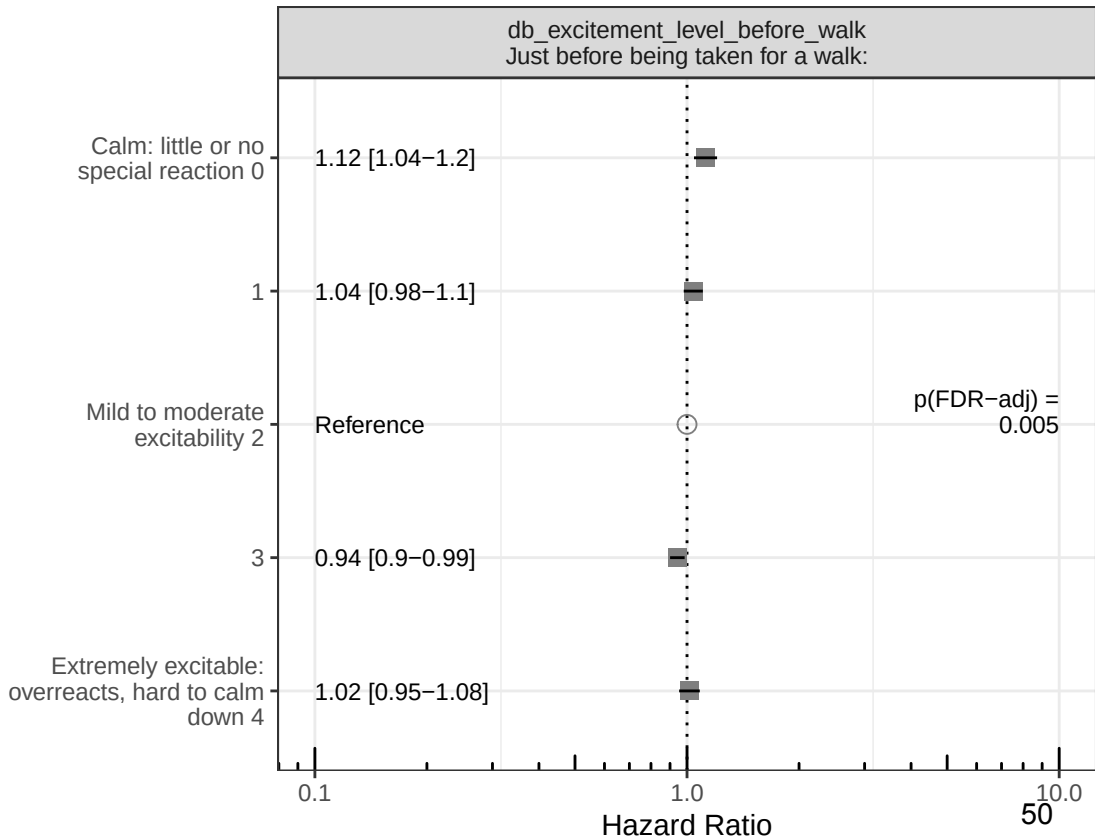
hs\_congenital\_condition\_present  
Was your dog born with a congenital disorder (defect present at birth)?



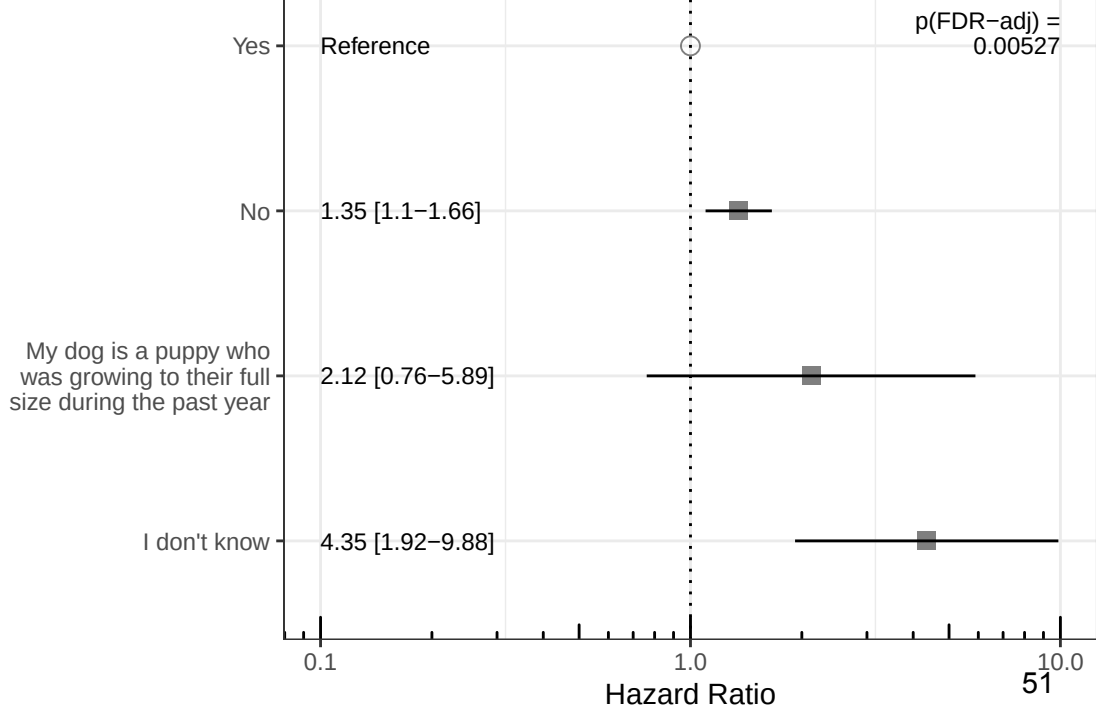


de\_wood\_fireplace\_present  
Wood burning fireplace or wood stove present:

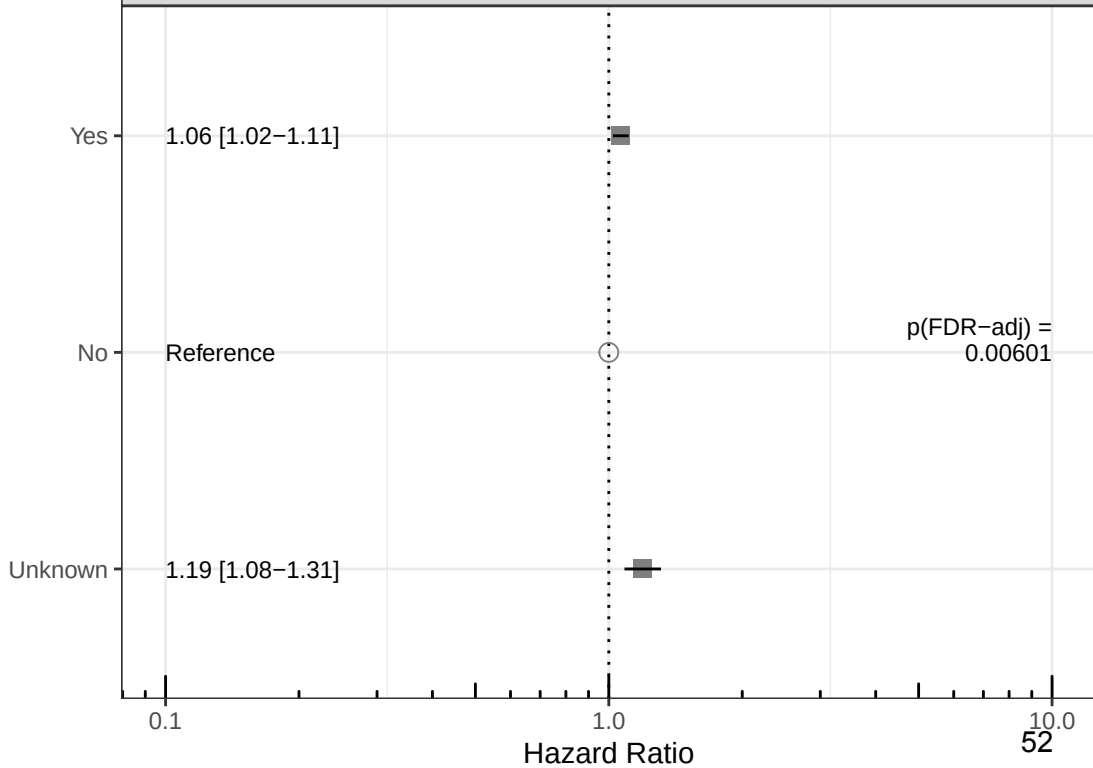




dt\_weight\_change  
In the past 12 months, has your dog's weight stayed about the same?



df\_ever\_overweight  
Has your dog ever been considered overweight?



hs\_recent\_hospitalization  
In the past three months, has your dog been hospitalized for more than 24 hours?

Yes 1.26 [1.12–1.42]

No Reference

p(FDR-adj) =  
0.0124

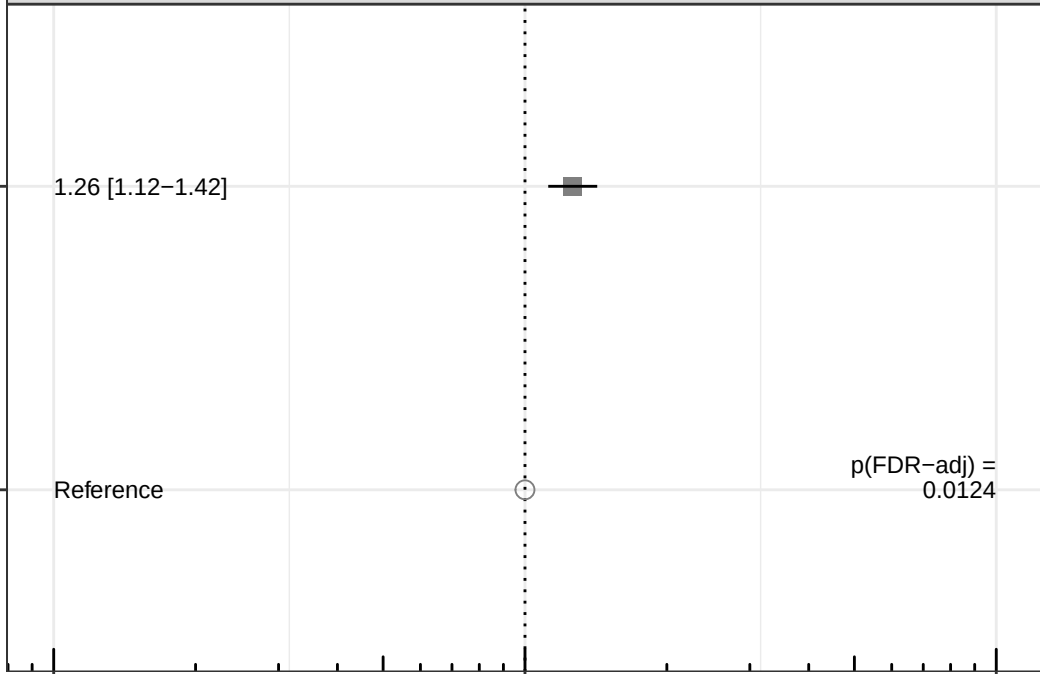
0.1

1.0

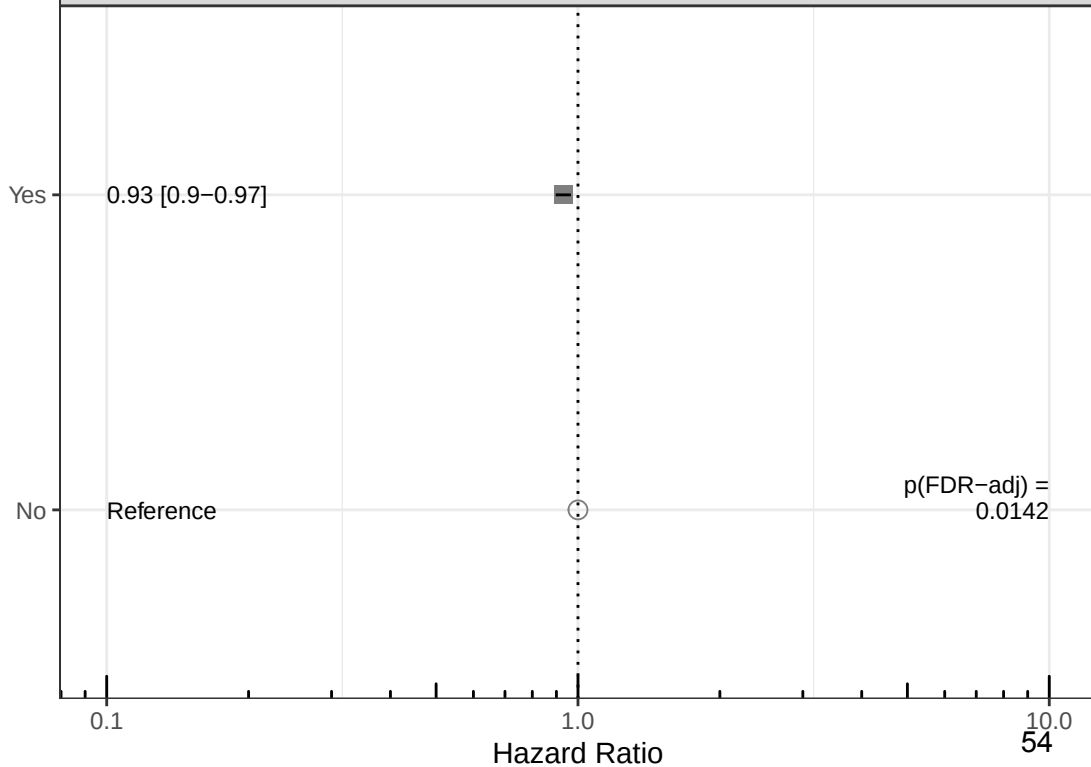
10.0

Hazard Ratio

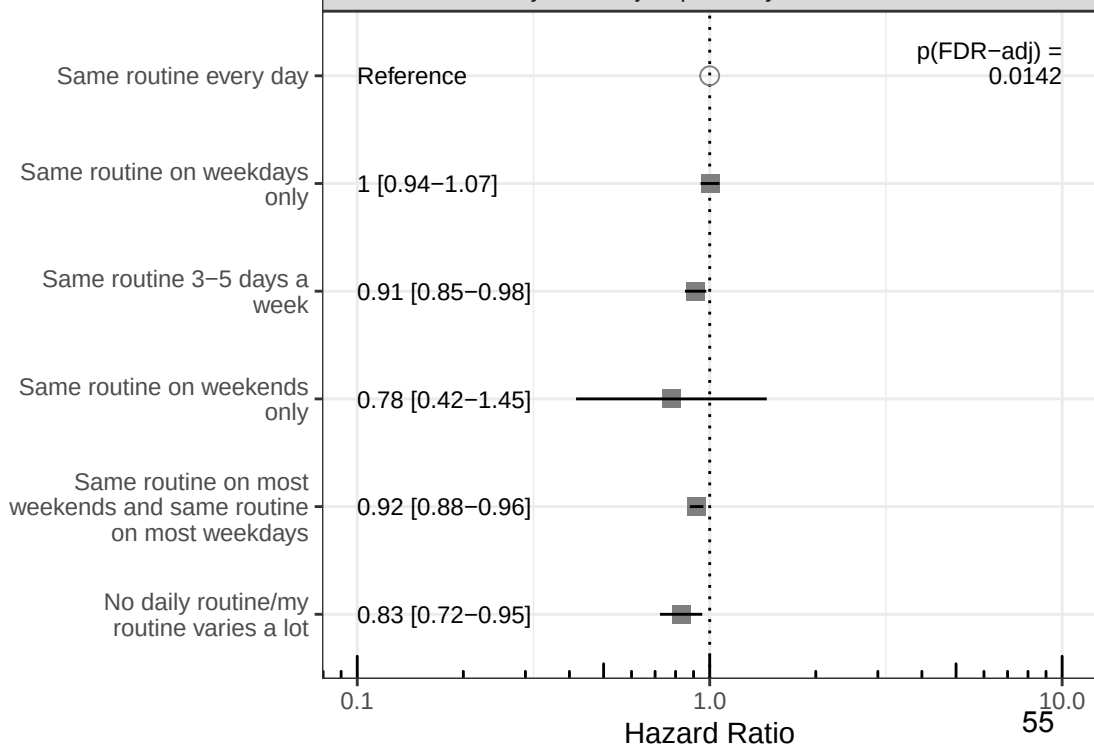
53



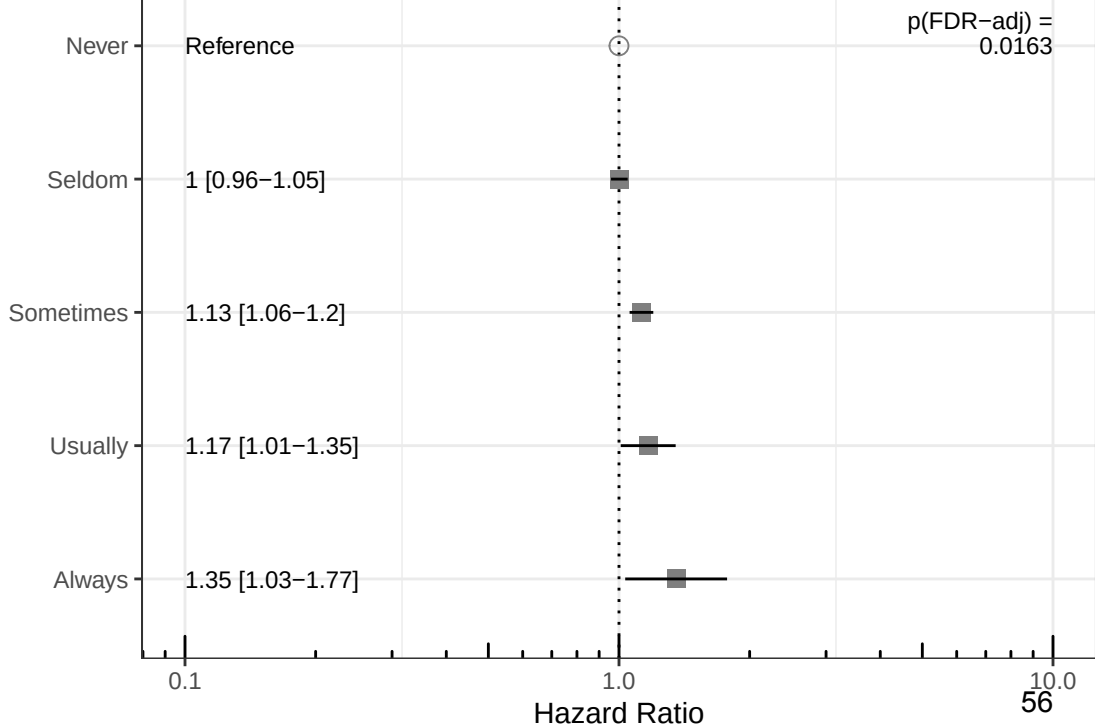
pa\_moderate\_weather\_outdoor\_other\_hard\_surface  
Hard surface other than concrete or wood (asphalt, pavement, street):



de\_routine\_consistency  
How consistent is your dog's daily routine? We understand that this can vary. Please just provide your best estimate.

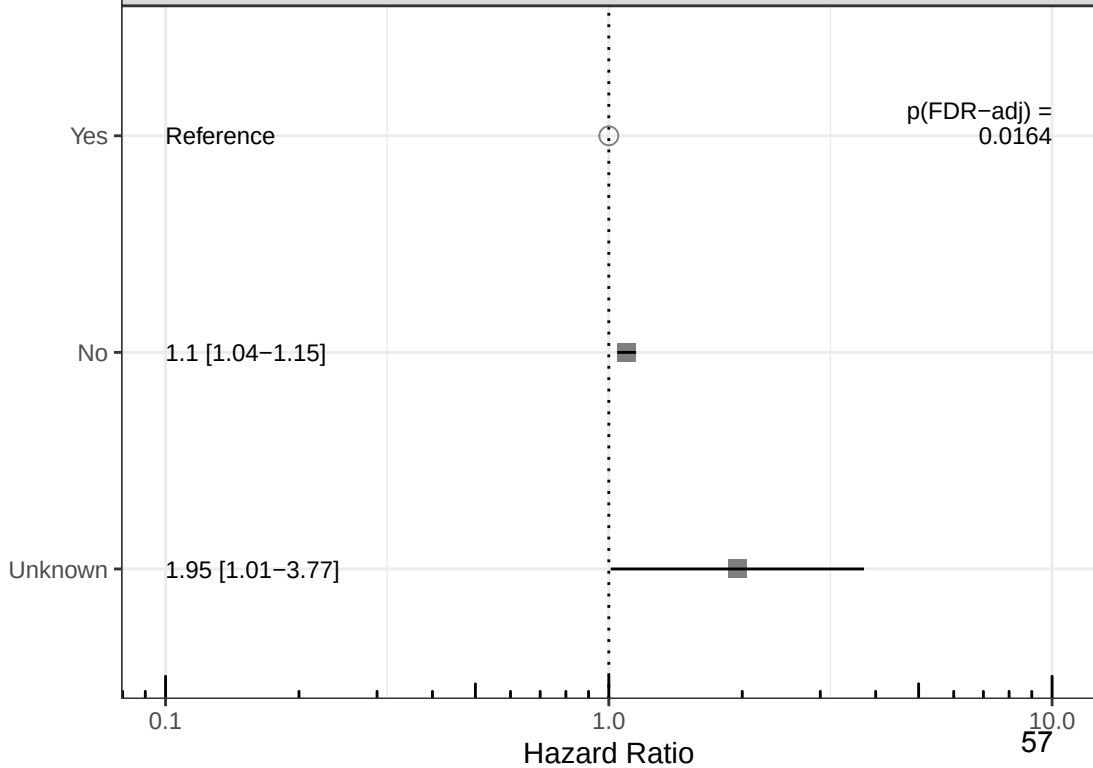


db\_hyperactive\_frequency  
Hyperactive, restless, has trouble settling down:





de\_routine\_toys\_include\_tennis\_balls  
Tennis ball



db\_aggression\_level\_delivery\_workers\_at\_home  
When letter carriers or other delivery workers approach your home:

No aggression: no visible  
signs of aggression 0

Reference

p(FDR-adj) =  
0.0194

1

0.95 [0.91–0.99]

Moderate aggression:  
growling/barking, baring  
teeth 2

0.92 [0.88–0.97]

3

0.83 [0.75–0.92]

Serious aggression:  
snaps, bites or attempts  
to bite 4

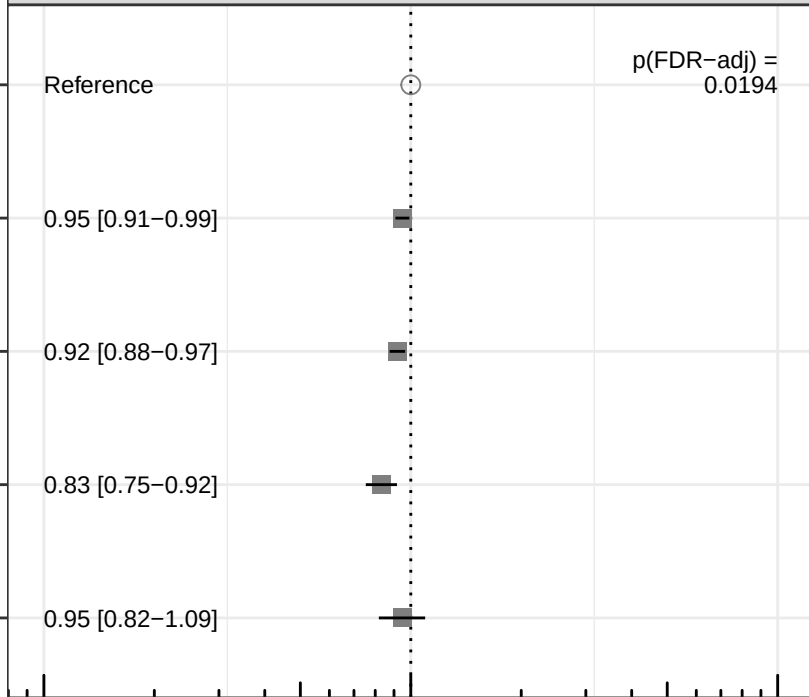
0.95 [0.82–1.09]

0.1

Hazard Ratio

10.0

58



df\_treats\_other  
Other

Yes 0.93 [0.89–0.96]

No Reference

p(FDR-adj) =  
0.0194

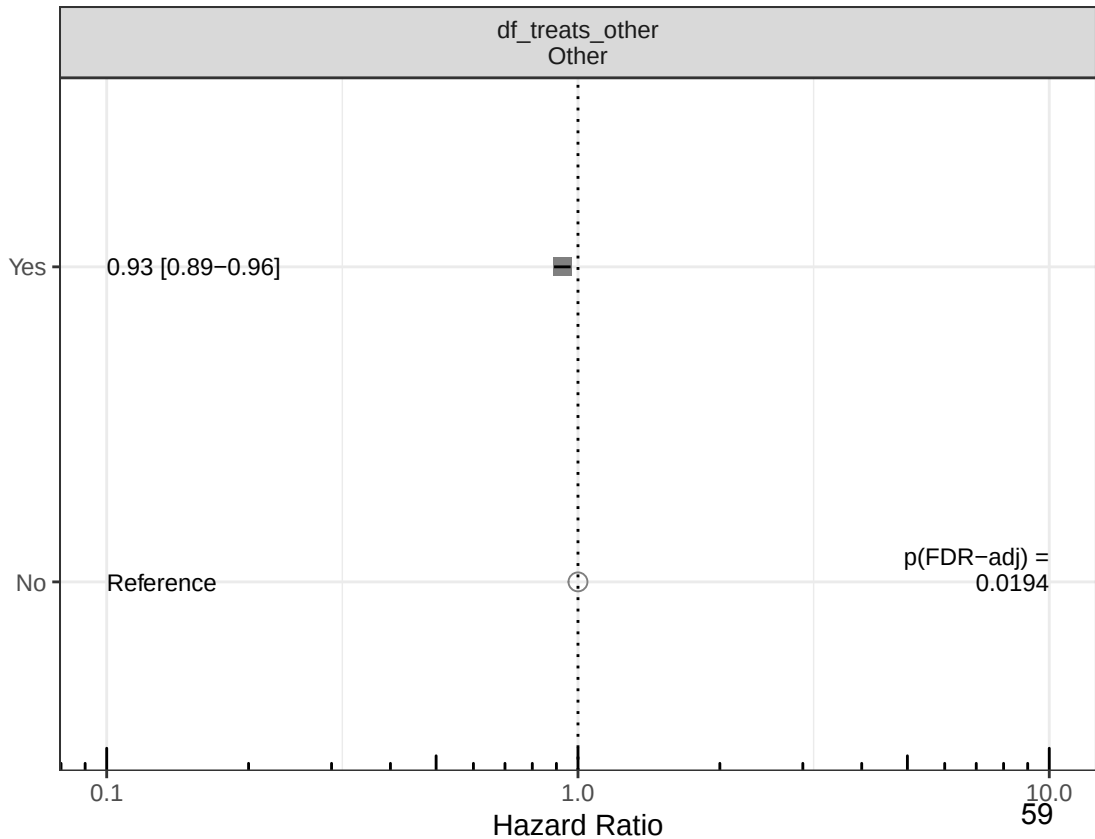
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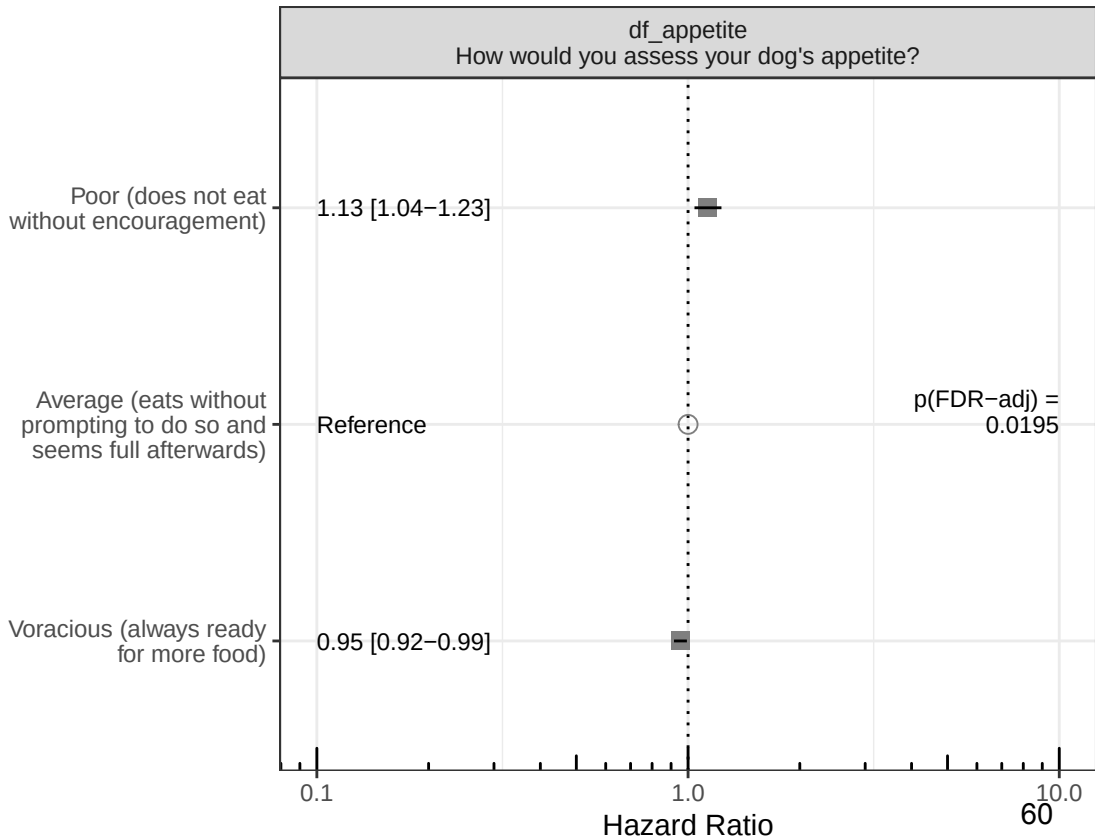
1.0

10.0

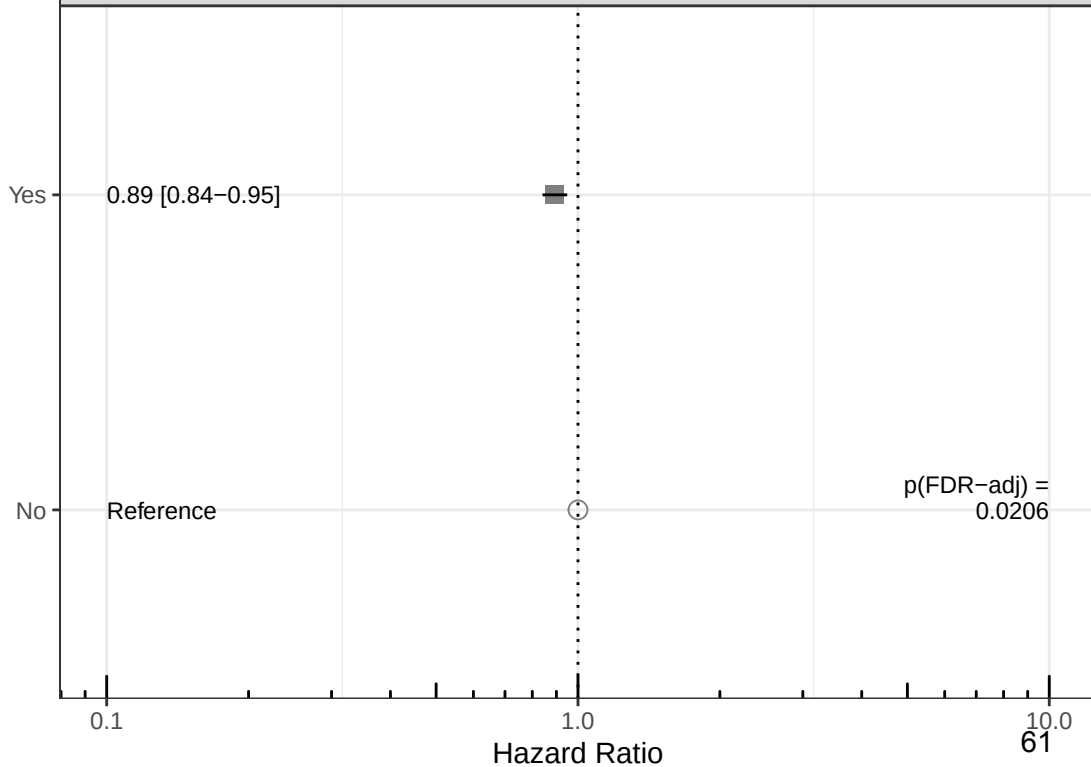
Hazard Ratio

59

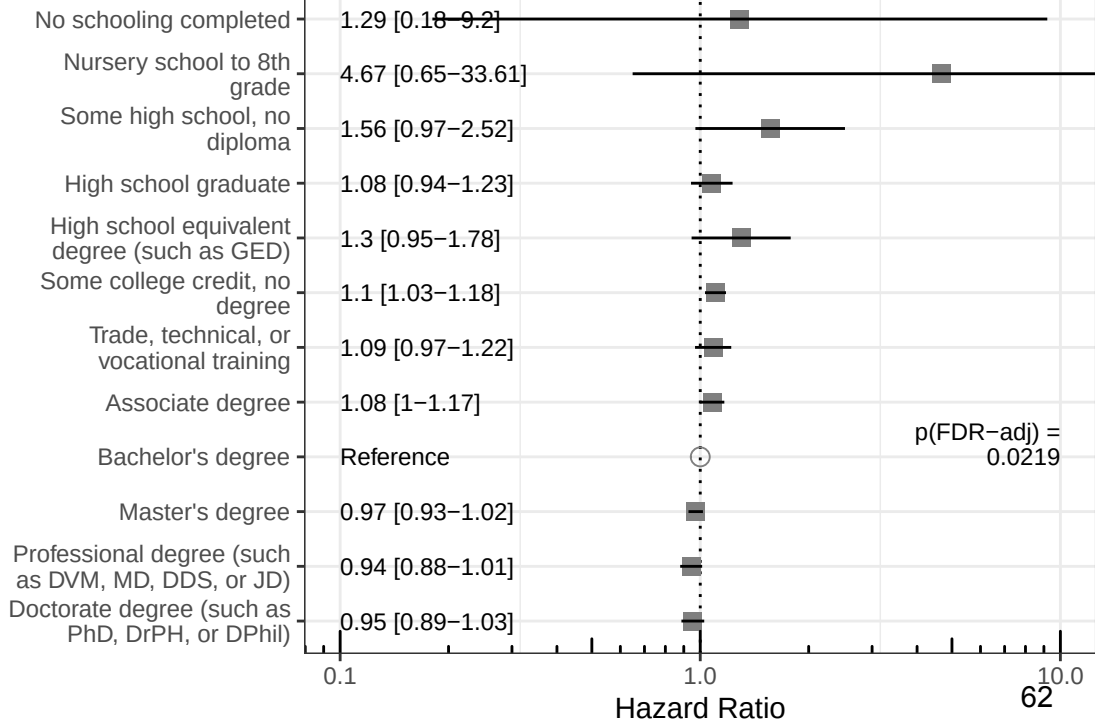




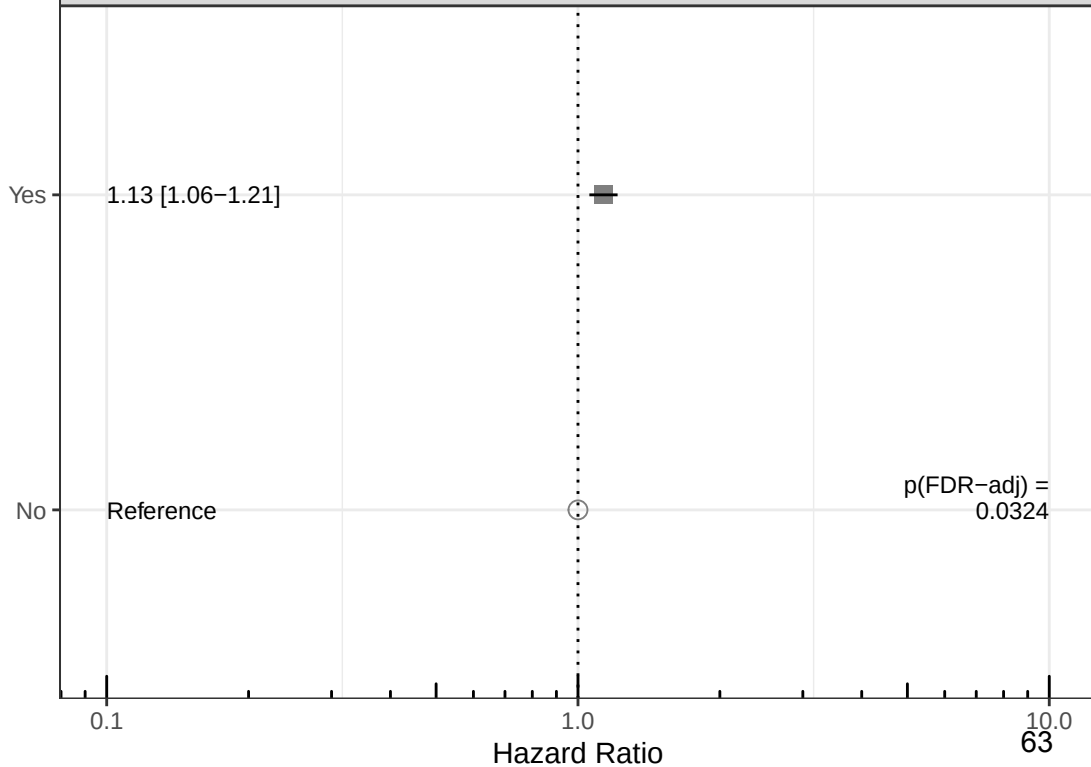
pa\_on\_leash\_walk\_reasons\_training\_obedience  
What is the reason for your activities? Training and obedience



od\_max\_education  
What is the highest degree or level of school you have completed? If currently enrolled, please indicate the highest degree received.



mp\_recent\_non\_prescription\_meds\_eye\_lubricant  
Given in past year: Eye lubricants



pa\_on\_leash\_walk\_frequency  
Over the past year, what is the average frequency that your dog is active on a lead/leash?

